

Seminar on Algebraic Geometry 3
Exposé VI A: Generalities on Algebraic Groups and Group Schemes
English Working Translation and PDF Reconstruction — Loop 1

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Checkpoint and reconstruction note

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Contents

Checkpoint and reconstruction note	i
Exposé VI-A. Generalities on Algebraic Groups	1
0. Preliminary Remarks	1
1. Local properties of an A -group locally of finite type	4
2. Connected components of an A -group locally of finite type	8
3. Construction of quotients $F \backslash G$ (for G, F of finite type)	22
4. Construction of quotients $F \backslash G$ (general case)	29
5. Connections with Exposé IV and consequences	33
6. Complements on k -groups not necessarily of finite type	39
Bibliography	42

Exposé VI_A

Generalities on Algebraic Groups

by P. Gabriel

⁰Throughout this chapter, A will denote an Artinian local ring with residue field k . A group scheme G over $\operatorname{Spec} A$ will be called simply an A -group. This A -group is said to be *locally of finite type* if its underlying scheme is locally of finite type over A ; it is said to be *algebraic* if its underlying scheme is of finite type over A .

0. Preliminary Remarks

0.1. Let us first consider a group scheme G over an arbitrary scheme S . We call the structural morphism

$$\mu : G \times_S G \longrightarrow G$$

multiplication, and we call the morphism $c : G \rightarrow G$, defined by $c(T)(x) = x^{-1}$ for every scheme T over S and every $x \in G(T)$, *inversion*. If U and V are subsets of the underlying set of G , we denote by $U \cdot V$ the image under multiplication of the subset of $G \times_S G$ consisting of those points whose first projection belongs to U and whose second projection belongs to V . Likewise, the notations U^{-1} and $c(U)$ are equivalent.

Let pr_1 be the projection of $G \times_S G$ onto the first factor, and let

$$\sigma : G \times_S G \longrightarrow G \times_S G$$

be the morphism with components pr_1 and μ . For every S -scheme T , the map $\sigma(T)$ is

$$(x, y) \mapsto (x, xy);$$

it follows that σ is an automorphism. The composite of this automorphism with the projection pr_2 of $G \times_S G$ onto the second factor is the multiplication morphism. When G is flat over S , pr_2 , and hence μ , are flat morphisms; when G is smooth over S , pr_2 , and hence μ , are smooth morphisms, and so on.

0.2. From now on we assume that S is the spectrum of an Artinian local ring A with residue field k . We denote by $(\mathbf{Sch}/k)_{\operatorname{red}}$ the category of reduced schemes over k . For every scheme X over A , the reduced scheme X_{red} is an object of $(\mathbf{Sch}/k)_{\operatorname{red}}$, and the functor

$$X \mapsto X_{\operatorname{red}}$$

is right adjoint to the inclusion of $(\mathbf{Sch}/k)_{\operatorname{red}}$ into $\mathbf{Sch}/A$. It follows that, for every A -group G , G_{red} is a group in the category $(\mathbf{Sch}/k)_{\operatorname{red}}$.¹ In other words, for every reduced k -scheme

⁰Editor's note: version of October 13, 2024.

¹Editor's note: the sentences that follow have been added.

T , $G_{\text{red}}(T)$ is endowed, functorially in T , with a group structure. One must bear in mind that G_{red} is not necessarily a k -group, because its “multiplication” is only a morphism

$$(G_{\text{red}} \times_k G_{\text{red}})_{\text{red}} \longrightarrow G_{\text{red}}^2.$$

⁶ However, if k is a perfect field, the inclusion of $(\mathbf{Sch}/k)_{\text{red}}$ into $\mathbf{Sch}/k$ commutes with products. Groups in $(\mathbf{Sch}/k)_{\text{red}}$ may therefore be identified with group schemes over k whose underlying schemes are reduced. In this case, if G is a k -group, G_{red} is a group subscheme of G ; in general, this subgroup is not normal in G .

For example, if k is a field of characteristic 3, the constant group $(\mathbb{Z}/2\mathbb{Z})_k$ acts nontrivially on the diagonalizable group $D_k(\mathbb{Z}/3\mathbb{Z})$ (cf. Exposé I, 4.1 and 4.4). If G denotes the semidirect product of $D_k(\mathbb{Z}/3\mathbb{Z})$ by $(\mathbb{Z}/2\mathbb{Z})_k$ defined by this action, then G_{red} is identified with $(\mathbb{Z}/2\mathbb{Z})_k$ and is not normal in G .⁷

Let k be an arbitrary field, let $k^{p^{-\infty}}$ be its perfect closure, and let H be a group in $(\mathbf{Sch}/k)_{\text{red}}$. Then

$$(H \otimes_k k^{p^{-\infty}})_{\text{red}}$$

is a group scheme over the perfect field $k^{p^{-\infty}}$. Since $H \otimes_k k^{p^{-\infty}}$ and $(H \otimes_k k^{p^{-\infty}})_{\text{red}}$ have the same underlying topological space, groups in $(\mathbf{Sch}/k)_{\text{red}}$ share with k -groups certain topological properties that are invariant under extension of the base field. For example, it will follow from 0.3 and the preceding remarks that every group in $(\mathbf{Sch}/k)_{\text{red}}$ is separated.

In what follows, we shall encounter groups in $(\mathbf{Sch}/k)_{\text{red}}$ whenever we deal with a nonempty locally closed subset U of an A -group G such that

$$U \cdot U = U \quad \text{and} \quad U^{-1} = U.$$

Indeed, the reduced subscheme of G defined by U is then a group in $(\mathbf{Sch}/k)_{\text{red}}$.

0.3. *An A -group G is always separated, because the unit section $e : \text{Spec } A \rightarrow G$ is a closed immersion.* Indeed, let x be the unique point of $\text{Spec } A$, and let $\eta : G \rightarrow \text{Spec } A$ be the structural morphism. Since $\eta \circ e = \text{id}_{\text{Spec } A}$, for every affine open $U = \text{Spec } B$ of G containing $e(x)$, the homomorphism $B \rightarrow A$ has a section and is therefore surjective. It follows that e is a closed immersion.⁸

The diagonal in $G \times_A G$ is identified with the functor from $(\mathbf{Sch}/A)^\circ$ to **Sets** which associates to every A -scheme S the inverse image of the identity element of $G(S)$ under the map

$$\varphi(S) : G(S) \times G(S) \longrightarrow G(S), \quad (x, y) \longmapsto x \cdot y^{-1}.$$

We therefore have the following cartesian square. Since the diagonal morphism is obtained from a closed immersion by base change, it is itself a closed immersion.

⁵Editor’s note: for examples of group schemes G over a nonperfect field k for which G_{red} is not a group scheme over k , see 1.3.2 below.

⁶Editor’s note: the remainder of 0.2, as well as 0.3, has been slightly modified.

⁷Editor’s note: for an analogous example with G connected, one may take, when $\text{char}(k) = p > 0$, the semidirect product of $\alpha_{p,k}$ and $\mathbb{G}_{m,k}$.

⁸Editor’s note: this argument is valid for every local ring A of dimension zero, and shows that, if S is a discrete scheme, every S -group is separated; cf. VI_B, 5.2. On the other hand (cf. VI_B, 5.6.4), if S contains a nonisolated closed point s , the S -scheme G obtained by gluing two copies of S along the open subset $S - \{s\}$ is not separated over S , although it carries an S -group structure.

$$\begin{array}{ccc}
G \times_S G & \xrightarrow{\varphi} & G \\
\uparrow \text{morphisme diagonal} & & \uparrow \text{section unité} \\
G & \longrightarrow & \text{Spec } A.
\end{array}$$

0.4.⁹ Let G be an A -scheme. We shall say that a point g of G is *strictly rational over A* if there exists an A -morphism

$$s : \text{Spec } A \longrightarrow G$$

which sends the unique point of $\text{Spec } A$ to g ; that is, if the homomorphism $A \rightarrow \mathcal{O}_{G,g}$ admits a retraction. One then has $\kappa(g) = k$, so g is a closed point of G . Indeed, if B is the ring of an affine open neighborhood of g , and if $\mathfrak{p}$ is the prime ideal of B corresponding to g , then $B/\mathfrak{p} \subset k$ is a finite A -algebra, hence an integral Artinian ring, and therefore a field.

Suppose henceforth that G is an A -group. Such a morphism $s : \text{Spec } A \rightarrow G$ defines an automorphism r_s of the scheme G over A , called *right translation by s* . For every morphism $\pi : S \rightarrow \text{Spec } A$, the automorphism $r_s(\pi)$ of $G(S)$ is defined by

$$x \longmapsto x \cdot G(\pi)(s), \quad x \in G(S).$$

Similarly, ℓ_s denotes *left translation by s* , namely the automorphism of G defined by

$$\ell_s(\pi)(x) = G(\pi)(s) \cdot x, \quad x \in G(S).$$

The schemes $G \otimes_A k$ and G have the same underlying topological space $\underline{G}$, $G \otimes_A k$ is a k -group, and $s \otimes_A k$ depends only on g , not on s . Consequently, the automorphisms of $\underline{G}$ induced by r_s and ℓ_s (or by $r_{s \otimes k}$ and $\ell_{s \otimes k}$) depend only on g , not on s . Thus, if P is a subset of $\underline{G}$, we may write

$$r_g(P) \text{ or } P \cdot g \quad (\text{respectively, } \ell_g(P) \text{ or } g \cdot P)$$

instead of $r_s(P)$ (respectively, $\ell_s(P)$), in agreement with 0.1.

Remark 0.4.1. *If g is a strictly rational point of G , and if $A \rightarrow A'$ is a morphism of Artinian local rings, then $G' = G \otimes_A A'$ has a unique point g' above g , and g' is strictly rational over A' . Moreover, if P' denotes the inverse image of P in G' , then $P' \cdot g'$ is the inverse image of $P \cdot g$ (cf. EGA I, 3.4.8).*

Proposition 0.5.¹⁰ *Let G be an A -group, and let U, V be two dense open subsets of G . Then $U \cdot V$, that is, the image of $U \times_A V$ under multiplication, is the entire underlying space of G .*

Proof. Since G and $G \otimes_A k$ have the same underlying space, after replacing A by k and G by $G \otimes_A k$, if necessary, we may assume that $A = k$. Let $g \in G$, and set $K = \kappa(g)$. Then left translation ℓ_g is an automorphism of G_K . Since the projection $G_K \rightarrow G$ is open, U_K and V_K are dense open subsets of G_K , as is the image of V_K under ℓ_g . Consequently there exists $v \in V_K$ such that $u = \ell_g(v)$ belongs to U_K .

⁹Editor's note: the original has been expanded in what follows; in particular, Remark 0.4.1 has been added.

¹⁰Editor's note: the original stated this result under the hypothesis that G be locally of finite type over A . Since the result is useful in the general case, and since the proof is essentially the same, it has been stated and proved here in full generality. It will be used several times below.

Let L be an extension of K containing $\kappa(v)$, and hence $\kappa(u)$, and let g_L and v_L be the L -points of G_L induced by g and v . Then

$$g_L \cdot v_L = u'$$

is a point of G_L above u , and therefore

$$g_L = u' \cdot v_L^{-1}$$

lies above $U \cdot V$. Thus $g \in U \cdot V$, proving the proposition. $\square$

Corollary 0.5.1.¹¹ *If G is an irreducible A -group, then G is quasi-compact.*

Proof. Let U be a nonempty affine open subset of G . Then U is dense in G , so by 0.5 the morphism

$$\mu : U \times_A U \longrightarrow G$$

is surjective. Hence G is quasi-compact, since $U \times_A U$ is. $\square$

Corollary 0.5.2.¹² *Let G be an A -group scheme, and let H be a group subscheme of G over A . Then H is closed.*

Proof. Let k' be the perfect closure of the residue field k of A . Since the underlying topological spaces of G and H are unchanged under the base change $A \rightarrow k \rightarrow k'$, we may assume that $A = k$ is a perfect field. We may then assume that G and H are reduced, hence geometrically reduced.

Let $\overline{H}$ be the closure of H . Then $\mu^{-1}(\overline{H})$ is a closed subset of $G \times G$ containing $H \times H$. The morphisms

$$H \longrightarrow \operatorname{Spec} k \quad \text{and} \quad \overline{H} \longrightarrow \operatorname{Spec} k$$

are universally open. Since H is dense in $\overline{H}$, $H \times H$ is dense in $\overline{H} \times H$, and $\overline{H} \times H$ is dense in $\overline{H} \times \overline{H}$. Thus $H \times H$ is dense in $\overline{H} \times \overline{H}$. Consequently

$$\mu(\overline{H} \times \overline{H}) \subset \overline{H}.$$

Since $\overline{H} \times \overline{H}$ is reduced, μ induces a morphism

$$\mu' : \overline{H} \times \overline{H} \longrightarrow \overline{H}.$$

Now let $g \in \overline{H}$, and put $K = \kappa(g)$. The projection $\overline{H}_K \rightarrow \overline{H}$ is open, so H_K and $\ell_g(H_K)$ are two dense open subsets of $\overline{H}_K$. Hence there exist $u, v \in H_K$ such that $\ell_g(v) = u$. As in the proof of 0.5, it follows that g belongs to $H \cdot H = H$. Therefore $\overline{H} = H$. $\square$

1. Local properties of an A -group locally of finite type

We shall first see that, if G is locally of finite type and flat over A , one can “make every closed point of G strictly rational” by means of a finite flat extension of the base.

1.1. Unless expressly stated otherwise, from now on we assume that G is an A -group locally of finite type. When A is a field k , we shall obtain in Exposé VII_B very precise results on the local rings of G .¹³ Here we content ourselves with a few elementary results.

¹¹Editor’s note: this corollary has been inserted here; cf. 2.4 and 2.6.2 below.

¹²Editor’s note: this corollary, used implicitly in VIII, 6.7, has been added; see also VI_B, 6.2.5.

¹³Editor’s note: for example, they are always complete intersections; see Exposé VII_B, 5.5.1. Moreover, if $\operatorname{char}(k) = 0$, then G is smooth (Exposé VI_B, 1.6.1; see also Exposé VII_B, 3.3.1).

Proposition 1.1.1. *Let x be a point of an A -group G locally of finite type and flat over A . Then the local ring $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and there is a system of parameters $a_1, \dots, a_n$ of $\mathcal{O}_{G,x}$ such that $\mathcal{O}_{G,x}/(a_1, \dots, a_n)$ is a finite flat A -module (hence finite free).*

We first suppose that A is equal to its residue field k . It then suffices to prove that $\mathcal{O}_{G,x}$ is Cohen–Macaulay, and we may restrict to the case in which x is a closed point (cf. EGA 0_{IV}, 16.5.13). By Lemma 1.1.2 below, G contains a closed point y such that $\mathcal{O}_{G,y}$ is Cohen–Macaulay. By SGA 1, I, § 9, this is equivalent to saying that, for every finite extension K of the base field k and every point $\bar{y}$ of $\bar{G} = G \otimes_k K$ lying above y , the local ring $\mathcal{O}_{\bar{G},\bar{y}}$ is Cohen–Macaulay. If K has been chosen sufficiently large, that is, if K contains a normal extension of k containing the residue fields $\kappa(y)$ and $\kappa(x)$, then $\bar{y}$ is (strictly) rational over K , as is every point $\bar{x}$ of $\bar{G}$ lying above x .¹⁴ Since the automorphism

$$r_{\bar{x}} \circ (r_{\bar{y}})^{-1}$$

sends $\bar{y}$ to $\bar{x}$, it follows that $\mathcal{O}_{\bar{G},\bar{x}}$, and hence $\mathcal{O}_{G,x}$ (SGA 1, I, § 9), are Cohen–Macaulay.

Now let A again be arbitrary. The preceding argument applies to $k \otimes_A G$, so that $k \otimes_A \mathcal{O}_{G,x}$ is Cohen–Macaulay. If $a_1, \dots, a_n$ is a sequence of elements of $\mathcal{O}_{G,x}$ whose image in $k \otimes_A \mathcal{O}_{G,x}$ is a system of parameters, it follows from SGA 1, IV, 5.7, or EGA 0_{IV}, 15.1.16, that $a_1, \dots, a_n$ is an $\mathcal{O}_{G,x}$ -regular sequence and that $\mathcal{O}_{G,x}/(a_1, \dots, a_n)$ is finite and flat (hence finite and free) over A .

Lemma 1.1.2. *Every nonempty scheme X , locally of finite type over an Artinian ring A , contains a closed point x whose local ring is Cohen–Macaulay.*

We may plainly suppose that X is affine, with algebra B , and argue by induction on $\dim X$ (the assertion is clear if X is discrete, since all its local rings are then Artinian). Since B is of finite type over A , if $\dim B > 0$, then B contains an element a that is not invertible and is not a zero divisor.¹⁵ The closed subscheme

$$X' = \operatorname{Spec} B/(a)$$

of X then has dimension strictly less than $\dim X$, and by the induction hypothesis it contains a closed point x such that $\mathcal{O}_{X',x}$ is Cohen–Macaulay. Since $\mathcal{O}_{X',x} = \mathcal{O}_{X,x}/(a)$, and a is neither invertible nor a zero divisor in $\mathcal{O}_{X,x}$, the ring $\mathcal{O}_{X,x}$ is Cohen–Macaulay (see also EGA IV₂, 6.11.3).

Proposition 1.2. *Let A be an Artinian local ring, let G be an A -group locally of finite type and flat over A , and let x be a closed point of G . Then there exists a local A -algebra A' , finite and free over A , such that every point of $G \otimes_A A'$ lying above x is strictly rational over A' .*

¹⁶ Indeed, let k_1 be a finite normal extension of k containing the residue field $\kappa(x)$ of x . By Lemma V.4.1.2, there exists a local A -algebra A_1 , finite and free over A , with residue field k_1 . In this case (cf. Editor’s note 14), all the points $g_1, \dots, g_n$ of $G \otimes_A A_1$ lying above $x \in G$ have k_1 as residue field; that is, $g_1, \dots, g_n$ are rational over A_1 in the sense of Exposé V, § 4.e).

¹⁴Editor’s note: indeed, the hypothesis on K implies that, for every extension L of K , every k -morphism $\kappa(x) \rightarrow L$ (respectively, $\kappa(y) \rightarrow L$) factors through K . Consequently, all the points of $G \otimes_k K$ lying above x or y have K as residue field; that is, they are (strictly) rational over K .

¹⁵Editor’s note: indeed, B is a Noetherian Jacobson ring (cf. [BAC], V, § 3.4). If every noninvertible element is a zero divisor, then every prime ideal is an associated prime ideal of B ; in particular, B has only finitely many maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$. Since B is a Jacobson ring, the intersection of the $\mathfrak{m}_i$ is the nilradical of B . It follows that each $\mathfrak{m}_i$ is a minimal prime ideal of B , and hence that $\dim B = 0$.

¹⁶Editor’s note: the original has been expanded in what follows.

Let $B_1, \dots, B_n$ be the local rings of $g_1, \dots, g_n$. By 1.1.1, the rings $B_1, \dots, B_n$ have quotients $B'_1, \dots, B'_n$ that are Artinian and finite and free over A_1 . Set

$$A' = B'_1 \otimes_{A_1} \cdots \otimes_{A_1} B'_n.$$

Then A' is local, finite, and free over A_1 , and, for each $i = 1, \dots, n$, there are surjective homomorphisms

$$B_i \otimes_{A_1} A' \twoheadrightarrow B'_i \otimes_{A_1} A' \twoheadrightarrow A',$$

the second induced by the multiplication map

$$B'_i \otimes_{A_1} B'_i \twoheadrightarrow B'_i.$$

Consequently, A' has the required property.

1.3. Let e be the unit element (or origin) of G , that is, the image of the unique point of $\text{Spec } A$ under the unit section $\text{Spec } A \rightarrow G$. By definition, e is strictly rational over A .

Proposition 1.3.1.¹⁷ *Let G be a group locally of finite type and flat over an Artinian ring A , and let K (respectively, $\bar{k}$) be the perfect closure (respectively, an algebraic closure) of the residue field k of A .*

(1) *For every closed point x of $\bar{G} = G \otimes_k \bar{k}$, the local rings $\mathcal{O}_{\bar{G},e}$ and $\mathcal{O}_{\bar{G},x}$ are isomorphic. In particular, the tangent spaces $T_e \bar{G}$ and $T_x \bar{G}$ have the same dimension.*

(2) *The following assertions are equivalent:*

- (i) $G \otimes_A K$ is reduced.
- (i bis) $\mathcal{O}_{G,e} \otimes_A K$ is reduced.
- (ii) G is smooth over A .
- (ii bis) G is smooth over A at the origin.

Proof. (1) Let x be a closed point of $\bar{G}$. There is exactly one $\bar{k}$ -morphism

$$s : \text{Spec } \bar{k} \rightarrow \bar{G}$$

whose image is x . Right translation r_s then induces an isomorphism from

$$\mathcal{O}_{\bar{G},e} = \mathcal{O}_{G,e} \otimes_k \bar{k}$$

onto $\mathcal{O}_{\bar{G},x}$, which proves (1).

We now prove (2). By SGA 1, II.2.1, we reduce at once to the case in which A is a field ($A = k$). The implications

$$(i) \Rightarrow (i \text{ bis}), \quad (ii) \Rightarrow (ii \text{ bis}), \quad (ii) \Rightarrow (i), \quad (ii \text{ bis}) \Rightarrow (i \text{ bis})$$

are evident, so it remains only to prove that (i bis) implies (ii).

Now (i bis) implies that $\mathcal{O}_{G,e} \otimes_k \bar{k}$ is reduced. Hence, by (1), $\mathcal{O}_{\bar{G},x}$ is reduced for every closed point x of $\bar{G}$, and therefore $\bar{G}$ is reduced. Since $\bar{G}$ is locally of finite type over $\bar{k}$, there is at least one closed point y such that $\mathcal{O}_{\bar{G},y}$ is regular. By (1), the local rings at all closed points of $\bar{G}$ are isomorphic to $\mathcal{O}_{\bar{G},e}$; thus all these local rings are regular. Consequently, $\bar{G}$ is smooth over $\bar{k}$, and hence G is smooth over k .

¹⁷Editor's note: part (1), which is useful in Examples 1.3.2 below, has been added.

¹⁸ We may now give the examples below, pointed out by M. Raynaud, of group schemes G over a nonperfect field k for which G_{red} is not a k -group scheme.

Examples 1.3.2. Let k be a nonperfect field of characteristic $p > 0$, let $t \in k - k^p$, let $\bar{k}$ be an algebraic closure of k , and let $\alpha \in \bar{k}$ satisfy $\alpha^p = t$.

- (1) Consider the additive group $\mathbb{G}_{a,k} = \text{Spec } k[X]$, and let G be the group subscheme, finite over k , defined by the additive polynomial $X^{p^2} - tX^p$. Then

$$G_{\text{red}} = \text{Spec } k[X]/X(X^{p(p-1)} - t)$$

is étale at the origin. If it were a group scheme over k , it would be smooth (by 1.3.1); but G_{red} is not geometrically reduced, so it is not a group scheme over k .

- (2) Consider

$$\mathbb{G}_{a,k}^4 = \text{Spec } k[X, Y, U, V],$$

and let G be the group subscheme defined by the ideal I generated by the additive polynomials

$$P = X^p - tY^p \quad \text{and} \quad Q = U^p - tV^p.$$

Then G is two-dimensional and irreducible, since

$$(G_{\bar{k}})_{\text{red}} \simeq \text{Spec } \bar{k}[Y, V]$$

is irreducible.

Let $A = k[X, Y, U, V]$, and let $\mathfrak{m}$ be its augmentation ideal. Denote by x, y, u, v the images of dX, dY, dU, dV in

$$\Omega_{A/k}^1 \otimes_A (A/\mathfrak{m}),$$

viewed as linear forms on the tangent space

$$k^4 = T_0 \mathbb{G}_{a,k}^4.$$

We shall show that the subspace $E = T_0 G_{\text{red}}$ equals k^4 . Otherwise, there would be a linear form

$$f = ax + by + a'u + b'v, \quad a, b, a', b' \in k,$$

not all of whose coefficients are zero, that vanishes on E . Recall that the formation of $\Omega_{A/k}^1$, and hence that of tangent spaces, commutes with base change (cf. EGA IV₄, 16.4.5), and identify f with its image in $(\bar{k}^4)^*$. Since

$$(G_{\bar{k}})_{\text{red}} \subset (G_{\text{red}})_{\bar{k}},$$

the form f vanishes on the subspace $T_0(G_{\bar{k}})_{\text{red}}$ of $\bar{k}^4$, which is defined by the equations

$$g_1 = x - \alpha y \quad \text{and} \quad g_2 = u - \alpha v.$$

Thus

$$f = \lambda g_1 + \mu g_2, \quad \lambda, \mu \in \bar{k}.$$

But $\lambda g_1 + \mu g_2$ belongs to k^4 only if $\lambda = \mu = 0$. This contradiction shows that $E = k^4$, and hence

$$T_0(G_{\text{red}})_{\bar{k}} = \bar{k}^4.$$

¹⁸Editor's note: the following sentence and Examples 1.3.2 have been added.

On the other hand, $R = XV - YU$ belongs to $\sqrt{I}$, since

$$R^p = (X^p - tY^p)V^p - Y^p(U^p - tV^p).$$

Consequently, the tangent space F at the point $(\alpha, 1, \alpha, 1)$ of $(G_{\text{red}})_{\bar{k}}$ is contained in the hyperplane H of $\bar{k}^4$ with equation

$$\alpha dV + dX - dU - \alpha dY = 0,$$

and therefore has dimension at most 3.¹⁹ It follows from part (1) of 1.3.1 that G_{red} is not a group scheme over k .

2. Connected components of an A -group locally of finite type

2.1. Let us first consider an arbitrary A -group G , and let G' be the connected component of the origin e in G . This connected component is plainly closed, so we may identify it with the reduced closed subscheme of G having G' as its underlying space.²⁰

Proposition 2.1.1. *For every extension K of the residue field k of A , the underlying space of $G' \otimes_A K$ is the connected component of the origin in the K -group $G \otimes_A K$; in other words, G' is geometrically connected.*

Indeed, let $(G \otimes_A K)'$ be the connected component of the origin in $G \otimes_A K$. Since the image of $(G \otimes_A K)'$ in G is connected and contains the unit element of G , that image is contained in G' . Thus $(G \otimes_A K)'$ is contained in the inverse image $G' \otimes_A K$ of G' in $G \otimes_A K$. The proposition therefore follows from the connectedness of $G' \otimes_A K$, which is proved in the following lemma.

Lemma 2.1.2. *Let X and Y be two connected schemes over a field k . If X contains a rational point, then $X \times_k Y$ is connected.*

We give a direct proof below of this result from EGA IV₂ (4.5.8 and 4.5.14).

Suppose first that Y is nonempty, connected, and affine, with algebra B . Then $X \times_k Y$ is the spectrum of the quasi-coherent $\mathcal{O}_X$ -algebra

$$\mathcal{B} = \mathcal{O}_X \otimes_k B.$$

We wish to show that every nonempty subset U of $X \times_k Y$ that is both open and closed equals $X \times_k Y$. Now U is affine over X , and its affine $\mathcal{O}_X$ -algebra is a direct factor of $\mathcal{B}$. It follows from Lemma 2.1.3 below that the image of U in X is open and closed, and hence is all of X . In particular, this image contains a rational point x of X , so U meets the inverse image of x in $X \times_k Y$. Since this inverse image is isomorphic to Y , and hence connected, U contains the whole inverse image. The same conclusion would hold for the complement of U in $X \times_k Y$ if U were different from $X \times_k Y$, which is absurd.

If Y is now an arbitrary k -scheme, the preceding argument shows that the fibers of the canonical projection

$$X \times_k Y \longrightarrow Y$$

are connected. If x is a rational point of X , all these fibers meet the subscheme $\{x\} \times_k Y$, which is itself connected. This proves the assertion.

¹⁹Editor's note: in fact, it is easy to see that $\sqrt{I}$ is generated by P, Q, R at every point other than 0, so $F = H$.

²⁰Editor's note: we shall see below (2.2) that G' is a group in the category $(\mathbf{Sch}_k)_{\text{red}}$.

Lemma 2.1.3. *Let X be a scheme, and let $\mathcal{A}$ be a quasi-coherent $\mathcal{O}_X$ -algebra that is locally²¹ a direct factor of a free $\mathcal{O}_X$ -module. Then the image of $\mathrm{Spec} \mathcal{A}$ in X is open and closed.*

Let V be this image. It is clear that V is contained in the support of $\mathcal{A}$. Conversely, if x belongs to the support of $\mathcal{A}$, then $\mathcal{A}_x$ is nonzero and is a free $\mathcal{O}_{X,x}$ -module, since, by Kaplansky, every projective module over a local ring is free. Consequently, the fiber of $\mathrm{Spec} \mathcal{A}$ at x , which is affine with algebra

$$\mathcal{A}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x),$$

is nonempty. Thus the image of $\mathrm{Spec} \mathcal{A}$ coincides with the support of $\mathcal{A}$.

If s is the unit section of $\mathcal{A}$, the equality $s_x = 0$ implies that s , and hence $\mathcal{A}$, vanish in a neighborhood of x . Therefore the support of $\mathcal{A}$ is closed. On the other hand, we may assume that $\mathcal{A}$ is a direct factor of the direct sum $\mathcal{L}$ of a family $(\mathcal{O}_X^\alpha)$ of copies of $\mathcal{O}_X$.²² For each β , let φ^β denote the restriction to $\mathcal{A}$ of the canonical projection

$$\mathcal{L} = \bigoplus_{\alpha} \mathcal{O}_X^\alpha \longrightarrow \mathcal{O}_X^\beta.$$

For $x \in X$, let $\mathcal{A}_x^*$ denote the dual module

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}).$$

Since $\mathcal{A}_x$ is a direct factor of $\mathcal{L}_x$, the restriction map

$$\mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{L}_x, \mathcal{O}_{X,x}) \longrightarrow \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{A}_x, \mathcal{O}_{X,x}) = \mathcal{A}_x^*$$

is surjective. If $\mathcal{A}_x \neq 0$, then, since $\mathcal{A}_x$ is free and nonzero, there exist $a \in \mathcal{A}_x$ and a linear form $\varphi \in \mathcal{A}_x^*$ such that $\varphi(a) = 1$. There is therefore a family (ξ^α) of elements of $\mathcal{O}_{X,x}$ such that, for every $u \in \mathcal{A}_{X,x}$,

$$\varphi(u) = \sum_{\alpha} \xi^\alpha \varphi^\alpha(u);$$

this sum is finite because $\varphi^\alpha(u) = 0$ except for finitely many α . Applying this to $u = a$, we find $\alpha_1, \dots, \alpha_n$ such that

$$1 = \varphi(a) = \xi^{\alpha_1} \varphi^{\alpha_1}(a) + \dots + \xi^{\alpha_n} \varphi^{\alpha_n}(a).$$

There is then an open neighborhood U of x on which a and the ξ^{α_i} come from sections

$$\tilde{a} \in \mathcal{A}(U) \quad \text{and} \quad \tilde{\xi}^{\alpha_i} \in \mathcal{O}_X(U).$$

The equality

$$\sum_i \tilde{\xi}^{\alpha_i} \varphi^{\alpha_i}(\tilde{a}) = 1 \quad \text{on } U$$

shows that $\tilde{a}_y \neq 0$ for every $y \in U$. (“The support of a projective module is open.”)

2.2. The notation remains that of 2.1. It is clear that G' is a reduced k -scheme. Lemma 2.1.2 shows that $G' \times_k G'$ is connected, so $(G' \times_k G')_{\mathrm{red}}$ is the reduced subscheme of $G \times_A G$ whose underlying space is the connected component of the origin. In particular, the multiplication morphism

$$\mu : G \times_A G \longrightarrow G$$

²¹Editor’s note: the word “locally” has been added. The reference to Kaplansky is *Projective modules*, Ann. of Math. **68** (1958), 372–377; see also [BAC], § II.3, Ex. 3.

²²Editor’s note: the original has been expanded in what follows.

induces a morphism

$$\mu' : (G' \times_k G')_{\text{red}} \longrightarrow G'$$

that makes G' a group in $(\mathbf{Sch}/k)_{\text{red}}$.

2.2.bis.²³ Recall (cf. Exposé V) that, if P is a scheme, $\underline{P}$ denotes the underlying topological space of P . We define an A -subfunctor G^0 of G by setting, for every A -scheme S ,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G'}\}.$$

Let $c : G \rightarrow G$ be the inversion morphism. Since $c(\underline{G'}) = \underline{G'}$, we have $c \circ u \in G^0(S)$ for every $u \in G^0(S)$. On the other hand, if $u, v \in G^0(S)$, then $u \boxtimes v$ sends $\underline{S}$ into the subspace of $G \times_A G$ consisting of the points whose two projections belong to $\underline{G'}$. This subspace is identified with the underlying space of $G' \times_A G'$, which is connected by Lemma 2.1.2. Consequently, $\mu \circ (u \boxtimes v)$ sends $\underline{S}$ into $\underline{G'}$. This proves that G^0 is a group subfunctor of G over A .

If the connected component of e is open in G , then the subfunctor G^0 is represented by the subscheme induced by G on this open subset, which is therefore a group subscheme of G ; this subscheme will also be denoted by G^0 . In this case, with the notation of 2.1,

$$G' = (G^0)_{\text{red}},$$

and the topological spaces $\underline{G'}$ and $\underline{G^0}$ coincide.²⁴

2.3. In accordance with our convention in 1.1, we again assume from now on that G is locally of finite type over A . Then G is locally Noetherian, hence locally connected;²⁵ thus every connected component of G is open.

We shall write G^0 for the subscheme induced by G on the connected component G' of the unit element. By 2.2.bis, G^0 is a group subscheme of G , which we shall call the *neutral component* of G . For every A -scheme S , therefore,

$$G^0(S) = \{u \in G(S) \mid u(\underline{S}) \subset \underline{G^0} = \underline{G'}\}.$$

Let G^α be an arbitrary connected component of G , and let

$$\nu^\alpha : G^\alpha \times_A G^0 \longrightarrow G$$

be the morphism defined by

$$\nu^\alpha(S)(g, \gamma) = g\gamma g^{-1}$$

for every $S \in (\mathbf{Sch}/A)$, $g \in G^\alpha(S)$, and $\gamma \in G^0(S)$. If e is the origin of G , the restriction of ν^α to $G^\alpha \times_A \{e\}$ is the trivial morphism. Since $G^\alpha \times_A G^0$ is connected by 2.1.2, it follows that ν^α factors through G^0 . Thus, for every A -scheme S , $G^0(S)$ is a normal subgroup of $G(S)$. We have proved the following proposition.²⁶

Proposition 2.3.1. *Let G be an A -group locally of finite type. Then the neutral component G^0 is an open normal group subscheme of G .*

Proposition 2.4. *Let G be an A -group locally of finite type.*

²³Editor's note: this paragraph has been added in order to make the connection with Exposé VI_B, § 3.

²⁴Editor's note: in this exposé, the notation G^0 is reserved for the case in which the connected component of e is open; in Exposé VI_B, § 3, this connected component will be denoted by $\underline{G^0}$ in every case. This is a slight abuse of notation, but it is compatible with the preceding discussion when the connected component is the underlying topological space of an open group subscheme G^0 of G .

²⁵Editor's note: indeed, in a Noetherian space there are only finitely many connected components, and each is therefore open; see also EGA I, 6.1.9.

²⁶Editor's note: the number 2.3.1 has been added in order to emphasize this statement. Moreover, G^0 is even a characteristic subgroup of G ; see 2.6.5(ii).

- (i) G^0 is irreducible, and $G^0 \otimes_A k$ is geometrically irreducible over k .
- (ii) G^0 is quasi-compact, and hence is of finite type over A .

²⁷ *Proof.* (i) Since G^0 and $G^0 \otimes_A k$ have the same underlying topological space, it suffices to prove the second assertion. Let $\bar{k}$ be an algebraic closure of k . By 2.2, $(G \otimes_A \bar{k})_{\text{red}}$ is a reduced $\bar{k}$ -group locally of finite type, and hence is smooth over $\bar{k}$ by 1.3.1. A fortiori, the local rings of $(G \otimes_A \bar{k})_{\text{red}}$ are integral domains. Therefore,²⁸ since $G \otimes_A \bar{k}$ is locally Noetherian, the connected components of $G \otimes_A \bar{k}$ are irreducible (cf. EGA I, 6.1.10). In particular, the connected component $G^0 \otimes_A \bar{k}$ (cf. 2.1.1) is irreducible.

(ii) We now show that G^0 is of finite type over A . Since G^0 is locally of finite type over A , it suffices to prove that G^0 is quasi-compact. Since G^0 is irreducible, this follows from 0.5.1.

Corollary 2.4.1. *Every connected component of G is irreducible,²⁹ of finite type over A ,³⁰ and of the same dimension as G^0 .*

Indeed, we may assume that A equals its residue field k . Let C be a connected component of G , let x be a closed point of C , let $\kappa(x)$ be the residue field of x , and let k' be a finite normal extension of k containing $\kappa(x)$. The canonical projection

$$\pi : C \otimes_k k' \longrightarrow C$$

is open and closed.³¹ Consequently, if C' is a connected component of $C \otimes_k k'$, the projection $C' \rightarrow C$ is surjective. Thus C' contains a point $y \in \pi^{-1}(x)$, and such a point is rational over k' (cf. the proof of 1.2); it follows that C' is the image of $G^0 \otimes_k k'$ under right translation r_y . Now $G^0 \otimes_k k'$ is of finite type over k' , by 2.4, and $\pi^{-1}(x)$ is finite, of cardinality at most $[k' : k]$. Consequently, $C \otimes_k k'$ is of finite type over k' , and hence C is of finite type over k .

Moreover, since $G^0 \otimes_k k'$ is irreducible by 2.4, the same is true of C' ,³² and hence also of C , because the projection $C' \rightarrow C$ is surjective.

Finally, we saw above that $C \otimes_k k'$ is the disjoint union of finitely many translates of $G^0 \otimes_k k'$. Since dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), it follows that C has the same dimension as G^0 . (Moreover, by EGA IV₂, 5.2.1, one has $\dim_g G = \dim G^0$ for every point $g \in G$.)

2.5. This paragraph has been added. The results that follow appear in Exposé VI_B, but could (or should) have appeared in VI_A; it is useful to have them available already, in order to make Theorem 3.2 below more precise.

Lemma 2.5.1. *Let $(A, \mathfrak{m})$ be an Artinian local ring, and let $k = A/\mathfrak{m}$ be its residue field.*

- (i) *If X is an A -scheme such that $X \otimes_A k$ is locally of finite type (respectively, of finite type) over k , then X has the same property over A .*
- (ii) *Let $u : X \rightarrow Y$ be a morphism of A -schemes. If $u \otimes_A k$ is an immersion (respectively, a closed immersion), then so is u .*

²⁷Editor's note: in the statement, " G^0 is geometrically irreducible" has been replaced by " $G^0 \otimes_A k$ is geometrically irreducible over k ," and the proof has been expanded.

²⁸Editor's note: the original has been expanded here, and the reference to EGA I, 6.1.10, has been added.

²⁹Editor's note: one should bear in mind that a nonneutral connected component is not geometrically connected in general. For example, if $k = \mathbb{R}$, the group $\mu_{3,\mathbb{R}}$, represented by $\mathbb{R}[X]/(X^3 - 1)$, has two connected components: $\{e\} = \text{Spec } \mathbb{R}$ and $C = \text{Spec } \mathbb{R}[X]/(X^2 + X + 1)$; and $C \otimes_{\mathbb{R}} \mathbb{C}$ has two components.

³⁰Editor's note: the assertion that follows has been added; cf. Exposé VI_B, 1.5.

³¹Editor's note: the original has been expanded in what follows.

³²Editor's note: the original has been simplified here.

Proof. (i) Suppose that $X \otimes_A k$ is locally of finite type over k . Let $U = \operatorname{Spec} B$ be an affine open subset of X . By hypothesis, there are elements $x_1, \dots, x_n$ of B whose images generate $B/\mathfrak{m}B$ as a k -algebra, and the “nilpotent Nakayama lemma” then shows that the x_i generate B as an A -algebra. This proves that X is locally of finite type over A . If, in addition, $X \otimes_A k$ is quasi-compact, then so is X , which has the same underlying topological space; hence X is of finite type over A . This proves (i).

We prove (ii). Suppose that $u \otimes_A k$ is an immersion (respectively, a closed immersion). Then u is a homeomorphism of X onto a locally closed (respectively, closed) subset of Y , and, for every $x \in X$, the ring homomorphism

$$\varphi_x : \mathcal{O}_{Y, u(x)} \longrightarrow \mathcal{O}_{X, x}$$

is such that $\varphi_x \otimes_A k$ is surjective. The nilpotent Nakayama lemma therefore shows that φ_x is surjective, so u is an immersion (respectively, a closed immersion).

Proposition 2.5.2. *Let A be an Artinian local ring with residue field k , and let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type.*

- (a) *The set $u(G)$ is closed in H , and its connected components are irreducible and all have the same dimension.*
- (b) *One has*

$$\dim G = \dim u(G) + \dim \operatorname{Ker}(u).$$

- (c) *If u is a monomorphism, then it is a closed immersion.*

Proof. By the preceding lemma, it suffices to prove the proposition when $A = k$. Moreover, the properties under consideration are stable under fpqc descent, and dimension is invariant under extension of the base field, so we may assume that k is algebraically closed.

We prove (a). Let C be the reduced subscheme of H whose underlying topological space is $u(G)$. Since $u(G)$ is stable under the inversion morphism of H , so is C . Furthermore, $u : G \rightarrow C$ is quasi-compact and dominant. Thus, by EGA IV₂, 2.3.7, the same is true of

$$u \times_k \operatorname{id}_G \quad \text{and} \quad \operatorname{id}_H \times_k u,$$

and hence of their composite

$$u \times_k u : G \times_k G \longrightarrow C \times_k C.$$

Consequently, multiplication on H sends $C \times_k C$ into C , and therefore C is a group subscheme of H .

Replacing H by C , we are thus reduced to the case in which u is dominant. Then $G(k)$ is dense in H , and hence meets every connected component of H ; it therefore acts transitively on the set of those connected components. It is consequently enough to show that $u(G)$ contains H^0 . Replacing G by $u^{-1}(H^0)$, we may therefore suppose that $H = H^0$. In that case, 2.4 shows that H is irreducible and of finite type over k , and hence Noetherian. Moreover, u is locally of finite type (cf. EGA I, 6.6.6) and quasi-compact, and is therefore of finite type. By Chevalley’s constructibility theorem (cf. EGA IV₁, 1.8.5), $u(G)$ is consequently a constructible (and dense) subset of

$$H = \overline{u(G)},$$

and thus contains a dense open subset U of H (cf. EGA 0_{III}, 9.2.2). By 0.5, we then have

$$H = U \cdot U \subset u(G),$$

whence $u(G) = H$. Together with 2.4.1, this proves (a).

We prove (b). Recall first that the functor $\text{Ker}(u)$ (cf. I, 2.3.6.1) is represented by $u^{-1}(e)$, where e denotes the identity element of H . Since u is of finite type, $\text{Ker}(u)$ is of finite type over k . On the other hand, replacing H by the reduced closed subscheme $u(G)$, we may assume that u is surjective. Let u^0 denote the restriction of u to G^0 . Since G and $\text{Ker}(u)$ are equidimensional, and since

$$\text{Ker}(u)^0 \subset \text{Ker}(u^0),$$

we are reduced to the case in which G , and hence also H , is irreducible.

By EGA IV₃, 9.2.6.2 and 10.6.1(ii), the set of $y \in H$ such that

$$\dim u^{-1}(y) = \dim G - \dim H$$

contains a nonempty open subset V . Since u is surjective, $U = u^{-1}(V)$ is then a nonempty open subset of G , and therefore contains a closed point x of G , because G is a Jacobson scheme (cf. EGA IV₃, 10.4.8). Right translation r_x is an isomorphism from $\text{Ker}(u)$ onto $u^{-1}(u(x))$, whence

$$\dim \text{Ker}(u) = \dim u^{-1}(u(x)) = \dim G - \dim H.$$

We prove (c), following [DG70], I, § 3.4. (Another proof is given in Exposé VI_B, 1.4.2.) Suppose that u is a monomorphism. If C is a connected component of G , there is a closed point $x \in G$ such that $C = r_x(G^0)$. If u_C (respectively, u^0) denotes the restriction of u to C (respectively, to G^0), then

$$u_C = r_{u(x)} \circ u^0 \circ r_x^{-1}.$$

It therefore suffices to show that u^0 is a closed immersion. We may thus suppose that $G = G^0$, so that G is irreducible and of finite type over k .

Let ξ be the generic point of G . Then $\mathcal{O}_{G,\xi}$ is an Artinian local ring; denote its maximal ideal by $\mathfrak{m}$. Let $h = u(\xi)$, let $\mathfrak{n}$ be the maximal ideal of $\mathcal{O}_{H,h}$, and set

$$A = \mathcal{O}_{G,\xi} / \mathfrak{n} \mathcal{O}_{G,\xi}.$$

Since u is a monomorphism, so is the morphism

$$u_h : \text{Spec}(A) \longrightarrow \text{Spec}(\kappa(h))$$

obtained by base change. The multiplication morphism

$$A \otimes_{\kappa(h)} A \longrightarrow A$$

is therefore an isomorphism (cf. EGA I, 5.3.8), whence $A = \kappa(h)$. Nakayama's lemma then shows that

$$\mathcal{O}_{H,h} \longrightarrow \mathcal{O}_{G,\xi}$$

is surjective, since $\mathfrak{n} \mathcal{O}_{G,\xi}$ is contained in $\mathfrak{m}$ and is therefore nilpotent.

Let V be an affine open subset of H containing h , let $U \neq \emptyset$ be an affine open subset of G contained in $u^{-1}(V)$, and let

$$\varphi : \mathcal{O}_H(V) \longrightarrow \mathcal{O}_G(U)$$

be the homomorphism of k -algebras induced by u . Let $\mathfrak{p}$ be the prime ideal of $\mathcal{O}_G(U)$ corresponding to ξ ,* and set $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Since G is of finite type over k , the k -algebra

*Translator's note: the French source prints η here; the generic point introduced immediately above is ξ .

$\mathcal{O}_G(U)$ is generated by finitely many elements $a_1, \dots, a_n$. By the preceding argument, there are elements $b_1, \dots, b_n, s \in \mathcal{O}_H(V)$, with $s \notin \mathfrak{q}$, such that in $\mathcal{O}_G(U)_{\mathfrak{p}}$ one has

$$a_i/1 = \varphi(b_i)/\varphi(s).$$

There are therefore elements

$$t_1, \dots, t_n \in \mathcal{O}_G(U) - \mathfrak{p}$$

such that

$$t_i(a_i\varphi(s) - \varphi(b_i)) = 0.$$

Set

$$t = t_1 \cdots t_n \varphi(s) \in \mathcal{O}_G(U) - \mathfrak{p}.$$

The equalities $a_i/1 = \varphi(b_i)/\varphi(s)$ then already hold in $\mathcal{O}_G(U)_t$. Since

$$t \in \mathcal{O}_G(U) = k[a_1, \dots, a_n],$$

there is an element $b \in \mathcal{O}_H(V)$ such that, for some $r \in \mathbb{N}$,

$$t/1 = \varphi(b)/\varphi(s)^r.$$

Consequently, φ induces a surjection

$$\mathcal{O}_H(V)_{sb} \longrightarrow \mathcal{O}_G(U)_t,$$

and hence u is a local immersion at ξ .

The open subset W of G consisting of the points at which u is a local immersion is therefore nonempty. Since G is a Jacobson scheme, W contains a closed point y . To prove that $W = G$, it is enough to show that every closed point of G belongs to W . But every closed point x is the image of y under the translation $r_x \circ r_y^{-1}$, and hence belongs to W . Thus $W = G$, which proves that u is a local immersion.

Since G is irreducible, it follows that u is an immersion. Indeed, for every $x \in G$, let U_x and V_x be open subsets of G and H , respectively, such that $x \in U_x$ and u induces a closed immersion of U_x into V_x . Since U_x is dense in G , the subset $u(U_x)$ is dense in $u(G) \cap V_x$. But $u(U_x)$ is closed in V_x , so

$$u(U_x) = V_x.$$

Moreover, since u is injective, $U_x = u^{-1}(V_x)$. Thus u induces a closed immersion of G into the open subscheme of H covered by the V_x . Hence $u : G \rightarrow H$ is an immersion. We have already seen that $u(G)$ is closed in H , so u is a closed immersion.

Lemma 2.5.3.³⁰ *Let A be an Artinian local ring, let k be its residue field, let G be a flat A -group, and let X be an A -scheme equipped with a left action*

$$\mu : G \times_A X \longrightarrow X$$

of G and with a section $s_0 : \operatorname{Spec} A \rightarrow X$. (This is the case, for example, if G' is a second A -group and one is given a morphism of A -groups $G \rightarrow G'$.)

Let φ be the morphism

$$\mu \circ (\operatorname{id}_G \times s_0) : G = G \times_A A \longrightarrow X.$$

³⁰Editor's note: no finiteness hypotheses are made in this statement; this will be useful later (cf. 6.2 and Exposé VI_B, § 12).

If φ is flat at a point g of G , then φ is flat.

Proof. Since G is flat over A , the fiberwise criterion for flatness (EGA IV₃, 11.3.10.2) shows that it is enough to prove that $\varphi \otimes_A k$ is flat. We may therefore suppose that $A = k$. In that case, giving s_0 is equivalent to giving a k -point $x_0 \in X(k)$, and φ is the morphism $h \mapsto hx_0$.

Let $h \in G$; we show that φ is flat at h . Let K be an extension of k containing copies of both $\kappa(g)$ and $\kappa(h)$. We have a cartesian square

$$\begin{array}{ccc} G_K & \xrightarrow{\phi_K} & X_K \\ \downarrow & & \downarrow \\ G & \xrightarrow{\phi} & X \end{array}$$

in which the two vertical arrows are faithfully flat. Thus, by Exposé V, 7.4(i), φ_K is flat at every point $g' \in G_K$ lying over g ; and, in order to show that φ is flat at h , it is enough to show that φ_K is flat at one point h' lying over h . We are therefore reduced to the case in which g and h are rational.

Set $u = hg^{-1}$, and let ℓ_u (respectively, μ_u) be left translation on G (respectively, on X) defined by u . Since $\varphi \circ \ell_u = \mu_u \circ \varphi$, we obtain a commutative square

$$\begin{array}{ccc} \mathcal{O}_{G,h} & \xrightarrow{\sim} & \mathcal{O}_{G,g} \\ \uparrow & & \uparrow \\ \mathcal{O}_{X,hx_0} & \xrightarrow{\sim} & \mathcal{O}_{X,gx_0} \end{array}$$

whose horizontal arrows are isomorphisms. Since the morphism

$$\mathcal{O}_{X,gx_0} \longrightarrow \mathcal{O}_{G,g}$$

is flat, the morphism

$$\mathcal{O}_{X,hx_0} \longrightarrow \mathcal{O}_{G,h}$$

is flat as well.

Proposition 2.5.4. *Let A be an Artinian local ring, let k be its residue field, let G be an A -group locally of finite type, and let $X \neq \emptyset$ be an A -scheme locally of finite type equipped with a left action of G . Suppose that the morphism*

$$\varphi : G \times_A X \longrightarrow X \times_A X, \quad (g, x) \longmapsto (gx, x),$$

defined on underlying sets by the displayed rule, is surjective. Then:

- (i) *The connected components of X are of finite type, irreducible, and all of the same dimension.*
- (ii) *More precisely, let $\bar{k}$ be an algebraic closure of k , and let x be a closed point of $X \otimes_A \bar{k}$. Its stabilizer F is a closed group subscheme of $G \otimes_A \bar{k}$, and the dimension of the irreducible components of X is*

$$\dim G - \dim F.$$

In view of Lemma 2.5.1, we may suppose that $A = k$. Suppose first that k is algebraically closed. Then G_{red} is a k -group locally of finite type; hence, replacing G by G_{red} and X by X_{red} , we may suppose that G and X are reduced.

Since $G \times_k X$ is locally of finite type over k , the morphism φ is locally of finite type (cf. EGA I, 6.6.6(v)), and is therefore locally of finite presentation because $X \times_k X$ is locally Noetherian. Let x be a rational point of X . The morphism

$$\varphi_x : G \longrightarrow X$$

obtained from φ by base change is surjective and locally of finite presentation. If η is a maximal point of X , then $\mathcal{O}_{X,\eta}$ is a field, since X is reduced; consequently, φ_x is flat at every point of G lying over η . By Lemma 2.5.3, φ_x is therefore flat. Thus $\varphi_x : G \rightarrow X$ is faithfully flat and locally of finite presentation, and hence open (cf. EGA IV₂, 2.4.6).

Since G^0 is open in G , irreducible, and quasi-compact (by 2.4), every orbit

$$G^0 x = \varphi_x(G^0),$$

as x ranges over the rational points of X , is an open, irreducible, and quasi-compact subset of X , and is therefore of finite type over k , since X is locally of finite type over k .

Every nonempty open subset of X contains a rational point, so these open subsets cover X . Furthermore, any two of them are either disjoint or equal. Indeed, if

$$\varphi_x(G^0) \cap \varphi_y(G^0)$$

is nonempty, it contains a rational point z . There are then rational points $g, h \in G^0$ such that

$$gx = z = hy,$$

whence $x = g^{-1}z$ and $y = h^{-1}z$, and consequently

$$\varphi_x(G^0) = \varphi_z(G^0) = \varphi_y(G^0).$$

It follows that the orbits $\varphi_x(G^0)$ are also closed, and are therefore at once the connected components and the irreducible components of X .

Finally, let x, y be two rational points of X . Since φ_x is surjective, there is a rational point $g \in G$ such that $y = gx$. Since G^0 is a normal subgroup of G , the orbit $G^0 y$ is the image of $G^0 x$ under translation ℓ_g on X ; hence $G^0 y$ and $G^0 x$ have the same dimension.

Moreover, by Exposé I, 2.3.3.1, the stabilizer of x is represented by the closed subscheme F of G defined by the following cartesian square:

$$\begin{array}{ccc} F & \longrightarrow & G \\ \downarrow & & \downarrow \phi_x \\ \text{Spec } k & \longrightarrow & X \end{array}.$$

Thus F is a k -group locally of finite type, while $F \cap G^0$ is a k -group of finite type containing F^0 . By 2.4.1, the schemes F and $F \cap G^0$ are equidimensional, of the same dimension as F^0 . Let

$$C = \varphi_x(G^0)$$

*Translator's note: the French source prints G^0 here; as the object being translated is an orbit in X , the intended source expression is $G^0 x$.

be the irreducible component of X containing x . Arguing as in the proof of part (b) of 2.5.2, we obtain

$$\dim C = \dim G^0 - \dim F^0 = \dim G - \dim F.$$

In the general case, that is, when k is an arbitrary field, let $\bar{k}$ be an algebraic closure of k . Let C be a connected component of X , and let C' be a connected component of $C \otimes_k \bar{k}$. Then C' is a connected component of

$$X' = X \otimes_k \bar{k}.$$

The morphism $\pi : X' \rightarrow X$ is open (cf. EGA IV₂, 2.4.10), and, since it is integral, it is also closed; therefore $\pi(C') = C$. Since C' is irreducible and quasi-compact, C is irreducible and quasi-compact, and hence is of finite type over k , because X is locally of finite type over k .

Finally, dimension is invariant under extension of the base field (cf. EGA IV₂, 4.1.4), so $\dim C = \dim C'$. Since all the irreducible components of X' have the same dimension, the same is true of those of X .

2.6. Complements.— This paragraph has been added; it is taken from [Per75], II, §§ 1–2, with complements due to O. Gabber.³¹ It shows that the preceding results are valid for every group scheme G over a field k . (This will be used in Sections 5, 6, and 7 of Exposé VI_B.)

Fix a field k . We begin with the following lemma (loc. cit., II, 2.1.1), which does not appear explicitly in EGA IV₂, § 4.4 (although it can probably be read between the lines at the beginning of loc. cit., § 4.4.1).

Lemma 2.6.0. *Let X be an irreducible k -scheme, let K be an extension of k , and let X' be an irreducible component of X_K . The projection $X' \rightarrow X$ is surjective.*

Proof. Let B be a transcendence basis of K over k , and set $L = k(B) \subset K$. By EGA IV₂, 4.3.2 and 4.4.1, X_L is irreducible and X' dominates X_L . Hence $X' \rightarrow X_L$ is integral and dominant, and therefore surjective. Since $X_L \rightarrow X$ is surjective (loc. cit., 4.4.1), so is $X' \rightarrow X$.

For the rest of 2.6, fix a k -group scheme G and an action $\mu : G \times X \rightarrow X$ of G on a k -scheme X , satisfying the following condition:

$$\Phi : G \times X \longrightarrow X \times X, \quad (g, x) \longmapsto (gx, x) \quad \text{is surjective.} \quad (\star)$$

(This holds in particular when G acts on itself by left translations.) For short, we shall say: “Let X be a G -scheme satisfying $(\star)$.” Finally, write k' for the perfect closure of k .

Proposition 2.6.1. *Let X be a G -scheme satisfying $(\star)$.*

- (i) *X is geometrically pointwise irreducible over k : for every extension K of k , each point $x \in X_K$ belongs to a unique irreducible component of X_K .*
- (ii) *Every local ring of $(X_{k'})_{\text{red}}$ is normal.*
- (iii) *Let η be a maximal point of $(X_{k'})_{\text{red}}$, put $C = \overline{\{\eta\}}$, and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme and is geometrically irreducible over L ; that is, $C \otimes_L K$ is irreducible for every extension K of L .*
- (iv) *In particular, if x is a rational point of X , then the irreducible component of X containing x is geometrically irreducible over k .*

³¹Editor’s note: these results were communicated to us by O. Gabber, in particular 2.6.6, which plays an important role in Section 5 of Exposé VI_B.

Proof. (i) Since hypothesis $(\star)$ is preserved by every base change $k \rightarrow K$, it is enough to show that every $x \in X$ belongs to a unique irreducible component of X . Since $\mathrm{Spec}(k') \rightarrow \mathrm{Spec}(k)$ is a universal homeomorphism, we may further suppose that k is perfect. We may then suppose that G and X are reduced.

Let η be a maximal point of X , and let z be any point of $Z = \overline{\{\eta\}}$. Since X is reduced, $\mathcal{O}_{X,\eta} = \kappa(\eta)$; and, since k is perfect, $\kappa(\eta) \otimes_k K$ is normal for every extension K of k (cf. EGA IV₂, 6.14.2). Thus every point of X_K lying above η is normal.

Since Φ is surjective, there is a point γ of $G \times X$ such that the two projections of $\Phi(\gamma)$ are z and η . Set $K = \kappa(\gamma)$. There are then rational points $g \in G_K$ and $\eta' \in X_K$, with η' lying above η , such that $z' = g\eta'$ lies above z . By the preceding paragraph, η' is a normal point of X_K , and therefore so is z' . Since $\pi : X_K \rightarrow X$ is flat, it follows that $z = \pi(z')$ is a normal point of X (cf. EGA IV₂, 2.1.13). This proves (ii) and (i).

The first assertion of (iii) now follows from (ii). Since L is algebraically closed in $\kappa(\eta)$, C is geometrically irreducible over L , by EGA IV₂, 4.5.9.

Finally, if X has a rational point x , it follows from (iii) that $L = k$, and hence the irreducible component C containing x is geometrically irreducible over k .^{*}This can also be seen directly as follows (cf. [Per75], II, 2.1). Let C be the irreducible component of X containing x , and let K be an extension of k . The scheme X_K has a unique point x_K above x , and by (i), x_K belongs to a unique irreducible component C' of X_K . On the other hand, by 2.6.0, every irreducible component of C_K contains x_K , and therefore equals C' .

Notation.— Write provisionally C^0 for the reduced closed subscheme of G whose underlying space is the unique irreducible component of G containing the identity element e .

Corollary 2.6.2. *C^0 is geometrically irreducible over k and is set-theoretically stable under the group law; that is, $(C_{k'}^0)_{\mathrm{red}}$ is a subgroup of $(G_{k'})_{\mathrm{red}}$. Consequently, C^0 is quasi-compact. *Proof.* By 2.6.1, C^0 is geometrically irreducible over k , so $C^0 \times C^0$ is irreducible and*

$$\nu(C^0 \times C^0) \subset C^0,$$

where ν denotes the morphism $(g, h) \mapsto gh^{-1}$. Since $\mathrm{Spec} k' \rightarrow \mathrm{Spec} k$ is a universal homeomorphism, the same conclusion holds for $C_{k'}^0$, and then for

$$H = (C_{k'}^0)_{\mathrm{red}}.$$

Since $H \times_{k'} H$ is reduced, ν induces a morphism

$$H \times_{k'} H \longrightarrow H;$$

thus H is a subgroup of $(G_{k'})_{\mathrm{red}}$. Consequently, by 0.5.1, H , and hence C^0 , is quasi-compact.

Recall 2.6.3.— Let Y be a scheme. Recall (EGA 0_{III}, 9.1.1) that a subset E of Y is called *retrocompact* if the inclusion $E \hookrightarrow Y$ is quasi-compact. Moreover, by EGA IV₁, 1.9.5(v) and 1.10.1, if U is a retrocompact open subset of Y , its closure $\overline{U}$ is the union of the closures $\overline{\{y\}}$ of the points $y \in U$. Of course, it is enough to let y run through the maximal points of U , that is, the maximal points of Y contained in U .

Consequently, if Y is quasi-separated and η is a maximal point of Y , then the intersection of the closed neighborhoods of η equals $\overline{\{\eta\}}$. Indeed, if $y \in Y - \overline{\{\eta\}}$, then y belongs to an affine open V not containing η . Since Y is quasi-separated, V is retrocompact, and therefore $\overline{V}$ is the union of the $\overline{\{\xi\}}$, for ξ running through the maximal points of Y belonging to V . Hence $\eta \notin \overline{V}$, so $Y - \overline{V}$ is a closed neighborhood of η not containing y .

Proposition 2.6.4. *Let X be a G -scheme satisfying $(\star)$, and let U be an open subset of X .*

^{*}Translator's note: the French source prints X in this conclusion and then switches from the point x to undefined symbols e and e_K . The statement and argument require the component C and the points x and x_K , respectively.

- (i) C^0U is an open subset of X , equal to the union of the irreducible components of X whose generic point belongs to U .
- (i') Consequently, if X is irreducible, it is quasi-compact.
- (ii) If, moreover, U is retrocompact in X , then $C^0U = \overline{U}$, and this subset is both open and closed in X .

Proof. (i) First, C^0U is open, since it is the union, for $g \in C^0$, of the projections of the open subsets

$$g \cdot U_{\kappa(g)} \subset X_{\kappa(g)},$$

and every projection $X_{\kappa(g)} \rightarrow X$ is open.

To prove the second assertion of (i), we may replace k by k' (since $\text{Spec } k' \rightarrow \text{Spec } k$ is a universal homeomorphism), and therefore suppose that k is perfect. We may then suppose that G and X are reduced, and hence geometrically reduced.

Let η be a maximal point of X contained in U , and put $Z = \overline{\{\eta\}}$. Consider

$$\mu' : C^0 \times Z \longrightarrow X.$$

Since C^0 is geometrically irreducible, $C^0 \times Z$ is irreducible; write γ for its generic point. The morphism μ' sends the point $\varepsilon(\eta)$, where ε denotes the identity section of C^0 , to η . Hence γ is sent to a generization of η , and therefore to η . Thus μ' sends the underlying space of $C^0 \times Z$ into Z , and, since $C^0 \times Z$ is reduced, μ' factors through Z .

Now let $z \in Z$, and put $K = \kappa(z)$. The morphism

$$\mu_z : G_K \longrightarrow X_K, \quad h \longmapsto h \cdot z,$$

is surjective. Let α be a maximal point of X_K . The local ring $\mathcal{O}_{X_K, \alpha}$ is a field, since X_K is reduced, so μ_z is flat at every point of G_K above α . It follows from Lemma 2.5.3 that μ_z is flat.

On the other hand, μ_z sends the generic point ω of C_K^0 to a point $t \in Z_K$. Let β be a maximal point of Z_K such that $t \in \overline{\{\beta\}}$. Since μ_z is flat, there is a generization ξ of ω such that $\mu_z(\xi) = \beta$. But ω is a maximal point of G_K , so necessarily $\xi = \omega$. Thus $\mu_z(\omega) = \beta$, and β lies above η , because $Z_K \rightarrow Z$ is flat.

Put $L = \kappa(\omega)$, and let ω_L and z_L be the L -points obtained from ω and z . Then

$$\omega_L \cdot z_L = \beta'$$

is a point of Z_L lying above β , and hence

$$z_L = \omega_L^{-1} \cdot \beta' \in C_L^0 \cdot U_L.$$

Therefore $z \in C^0 \cdot U$, which proves (i).

Since C^0 is quasi-compact by 2.6.2, assertion (i') follows: if X is irreducible and U is a nonempty affine open subset, then $X = C^0U$, so X is the image of the morphism $C^0 \times U \rightarrow X$, and is therefore quasi-compact. Finally, if U is retrocompact in X , then by 2.6.3, $\overline{U}$ is the union of the $\overline{\{\eta\}}$, where η runs through the maximal points of X contained in U . Hence $\overline{U} = C^0U$. This proves (ii).

One then obtains the following result ([Per75], II, Th. 2.4; see also [Per76], Prop. 4.1.1).

Theorem 2.6.5. *Let k be a field, and let G be a k -group scheme.*

- (i) *There is a unique group subscheme G^0 of G , called the identity component of G , such that:*

- (a) the underlying space of G^0 is the irreducible component of the identity element;
- (b) $G^0 \rightarrow G$ is a flat closed immersion; that is,

$$\mathcal{O}_{G,g} = \mathcal{O}_{G^0,g} \quad \text{for every } g \in G^0.$$

- (ii) Moreover, G^0 is quasi-compact, geometrically irreducible, and is a characteristic subgroup of G .
- (iii) If G is connected, then $G = G^0$.

Proof. (i) Recall first that G is separated (0.3), and hence, a fortiori, quasi-separated. Let U be an affine open subset of G containing the generic point ω of C^0 . By 2.6.3 and 2.6.4,

$$\overline{U} = C^0 U$$

is both open and closed, and, set-theoretically,

$$C^0 = \overline{\{\omega\}}$$

is the intersection of these open-and-closed subsets.

As U runs through the affine open subsets containing ω , the G -schemes $\overline{U}$ form a projective system whose transition morphisms are affine, since they are closed immersions. We can therefore form its projective limit G^0 (cf. EGA IV₃, 8.2.2). Thus, for every affine open V of G , $G^0 \cap V$ is the spectrum of the algebra

$$\lim_{\substack{\longrightarrow \\ U}} \mathcal{O}_G(V \cap \overline{U}) = \mathcal{O}_G(V) / \sum_U I_{\overline{U}}(V),$$

where $I_{\overline{U}}(V)$ denotes the kernel of

$$\mathcal{O}_G(V) \longrightarrow \mathcal{O}_G(V \cap \overline{U}).$$

It follows that the underlying space of G^0 is C^0 , and that $G^0 \rightarrow G$ is a closed immersion.

Moreover, for every $g \in G^0$, $\mathcal{O}_{G^0,g}$ is the inductive limit, as V runs through the affine open subsets of G containing g , of the k -algebras

$$\mathcal{O}_{G^0}(V \cap G^0) = \lim_{\substack{\longrightarrow \\ U}} \mathcal{O}_G(V \cap \overline{U}).$$

This double inductive limit is identified with

$$\lim_{\substack{\longrightarrow \\ U}} \lim_{\substack{\longrightarrow \\ g \in V \subset \overline{U}}} \mathcal{O}_G(V) = \mathcal{O}_{G,g}.$$

Thus

$$\mathcal{O}_{G^0,g} = \mathcal{O}_{G,g}$$

(see also EGA IV₂, 5.13.3(ii)), and consequently $i : G^0 \rightarrow G$ is a flat closed immersion. Conversely, this condition implies that $i^*(\mathcal{O}_G) = \mathcal{O}_{G^0}$, so G^0 is uniquely determined by conditions (a) and (b). This proves (i).

The first two assertions of (ii) follow from 2.6.2. Finally, let S be a k -scheme and let ϕ be an automorphism of the S -group G_S . For every $s \in S$, ϕ_s sends $G_{\kappa(s)}^0$ into itself, and therefore $\phi(G_S^0) \subset G_S^0$. Moreover, the closed immersion

$$i_S : G_S^0 \hookrightarrow G_S$$

obtained from i by base change is flat, so

$$\mathcal{O}_{G_S, \phi(z)} = \mathcal{O}_{G_S^0, \phi(z)} \quad \text{for every } z \in G_S^0.$$

Hence $\phi \circ i_S$ factors through G_S^0 . This proves that G^0 is a characteristic subgroup of G , and proves (ii). Finally, (iii) is a special case of part (i) of the following proposition.³²

Proposition 2.6.6. *Let k be a field, and let G be a k -group acting on a k -scheme X in such a way that*

$$G \times X \longrightarrow X \times X, \quad (g, x) \longmapsto (gx, x)$$

is surjective. Suppose that X is quasi-separated. Then:

- (i) *Every connected component C of X is irreducible.*
- (ii) *Let η be the generic point of C , and let L be the algebraic closure of k in $\kappa(\eta)$. Then C is an L -scheme, is geometrically irreducible over L , and the morphism*

$$G_L^0 \times_L C \longrightarrow C \times_L C$$

is surjective.

- (iii) *In particular, if C contains a rational point x , then C is geometrically irreducible over k , and*

$$\phi : G^0 \longrightarrow C, \quad g \longmapsto gx,$$

is surjective.

Proof. (i) Let C be a connected component of X , let Y be an irreducible component of X contained in C , let η be the generic point of Y , and let U be an affine open subset of X containing η . Since X is quasi-separated, U is retrocompact in X . Hence, by 2.6.4,

$$\overline{U} = G^0 U$$

is an open-and-closed subset of X which meets C , and therefore contains C . By 2.6.3, the intersection of the $\overline{U}$, as U runs through a fundamental system of affine open neighborhoods of η , is $\overline{\{\eta\}}$. It follows that

$$C = \overline{\{\eta\}},$$

which proves (i).

The first assertion of (ii), and also that of (iii), follows from 2.6.1. We first prove the second assertion of (iii). Let ω , respectively η , denote the generic point of G^0 , respectively C . Let $z \in C$, and set $K = \kappa(z)$. Since G^0 , respectively C , is geometrically irreducible over k , the scheme G_K^0 , respectively C_K , is irreducible; denote its generic point by ξ , respectively β . We saw in the proof of 2.6.4 that

$$\mu_z : G_K \longrightarrow C_K, \quad h \longmapsto h \cdot z,$$

sends ξ to β . Likewise, $\phi(\omega) = \eta$.

Put $L = \kappa(\xi)$, and let ξ_L and z_L be the L -points obtained from ξ and z . Then

$$\xi_L \cdot z_L = \beta'$$

is a point of C_L lying above $\beta \in C_K$, and hence also above $\eta \in C$. Consider the cartesian square

³²Editor's note: this proposition, as well as the preceding results, was communicated to us by O. Gabber. It will be used to correct the proof of Theorem 5.3 in Exposé VI_B.

$$\begin{array}{ccc} G_L^0 & \xrightarrow{\phi_L} & C_L \\ \pi_{G^0} \downarrow & & \downarrow \pi_C \\ G^0 & \xrightarrow{\phi} & C \end{array} ,$$

Since

$$\phi_L(\pi_{G^0}^{-1}(\omega)) = \pi_C^{-1}(\eta)$$

(cf. EGA I, 3.4.8), there is a $g \in G_L^0$ such that $\phi_L(g) = \beta'$. Therefore

$$z_L = \xi_L^{-1} \cdot \phi_L(g) = \phi_L(\xi_L^{-1}g),$$

whence

$$\phi(\pi_{G^0}(\xi_L^{-1}g)) = \pi_C(z_L) = z.$$

This proves that ϕ is surjective.

We now prove the second assertion of (ii). It is enough to show that, for every $z \in C$, the morphism

$$\mu_z : G_L^0 \otimes_L \kappa(z) \longrightarrow C \otimes_L \kappa(z)$$

is surjective. But this follows from (iii), since z is a rational point of $C \otimes_L \kappa(z)$.

Remark 2.6.7.— *Under the hypotheses of 2.6.6, if $C(k) = \emptyset$, the morphism*

$$G^0 \times_k C \longrightarrow C \times_k C$$

need not be surjective. For example, let $k = \mathbb{R}$ and $G = \{\pm 1\}_{\mathbb{R}}$. The G -torsor

$$X = \operatorname{Spec} \mathbb{R}[X]/(X^2 + 1)$$

is connected, but

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{R}} X$$

is not surjective. (Here $L = \mathbb{C}$, and

$$G^0 \times_{\mathbb{R}} X \longrightarrow X \times_{\mathbb{C}} X$$

is an isomorphism.)

3. Construction of quotients $F \backslash G$ (for G, F of finite type)

3.1.— Let A be an Artinian local ring, and let $u : F \rightarrow G$ be a homomorphism of A -groups. If

$$\mu : F \times_A F \longrightarrow F \quad \text{and} \quad \nu : G \times_A G \longrightarrow G$$

denote the multiplication morphisms, and if λ denotes the composite morphism

$$F \times_A G \xrightarrow{u \times G} G \times_A G \xrightarrow{\nu} G ,$$

recall that the *left quotient* $F \backslash G$ of G by F is the cokernel of the (Sch/ A)-groupoid G_* described below:

$$\begin{array}{ccccc}
& & \xrightarrow{F \times \lambda} & & \\
F \times_A F \times_A G & \xrightarrow{\mu \times G} & F \times_A G & \xrightarrow{\lambda} & G \\
& \xrightarrow{\text{pr}_{2,3}} & & \xrightarrow{\text{pr}_2} & \\
& & & &
\end{array}$$

Here pr_2 and $\text{pr}_{2,3}$ are the projections of $F \times_A G$ and $F \times_A (F \times_A G)$, respectively, onto the second factors. We shall say that G_* is the *groupoid with base G defined by u* (cf. Exposé V, § 2.a; as in Exposé V, we do not follow in this exposé the convention of IV, 4.6.15).

Since the unique A -morphism $F \rightarrow \text{Spec } A$ is universally open (EGA IV₂, 2.4.9), pr_2 is an open morphism. The same therefore holds for λ , which is the composite of pr_2 with the automorphism σ of $F \times_A G$ defined by

$$\sigma(S)(x, y) = (x, u(S)(x) \cdot y),$$

where S is a variable A -scheme and $x \in F(S)$, $y \in G(S)$. One sees in the same way that pr_2 and λ are flat when F is flat over A .

Let us also note, to conclude these preliminaries, that every A -morphism $s : \text{Spec } A \rightarrow G$ defines an automorphism of the groupoid G_* which induces on G , $F \times_A G$, and $F \times_A F \times_A G$, respectively, the automorphisms

$$r_s, \quad \text{id}_F \times_A r_s, \quad \text{id}_F \times_A \text{id}_F \times_A r_s.$$

We shall again write r_s for this automorphism of G_* , and shall say that r_s is the *right translation defined by s* (cf. 0.4).

Theorem 3.2. *Let F and G be flat groups locally of finite type over an Artinian local ring A . Let $u : F \rightarrow G$ be a quasi-compact homomorphism of A -groups whose kernel is finite over A . Then:*³³

- (i) *The left quotient $F \backslash G$ of G by F exists in (Sch/A) , and the sequence*

$$F \times_A G \xrightleftharpoons[\text{pr}_2]{\lambda} G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *The canonical morphism $p : G \rightarrow F \backslash G$ is surjective and open, and $F \backslash G$ is a direct sum of schemes of finite type over A .*
- (ii') *More precisely, $X = F \backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A , irreducible, and all of dimension $\dim G - \dim F$.*
- (iii) *The canonical morphism*

$$F \times_A G \longrightarrow G \times_{(F \backslash G)} G$$

is surjective.

- (iv) *If u is a monomorphism,*³⁴ *then:*

³³Editor's note: we have added (ii') and expanded point (iv), taking account of the additions made in 2.5.2, 2.5.4, and Exposé V, 6.1.

³⁴Editor's note: in this case, the hypothesis that G be flat can be omitted; cf. subsection 3.3.

(a) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \text{pr}_2)} G \times_{(F \setminus G)} G$$

is an isomorphism, and $G \rightarrow F \setminus G$ is faithfully flat and locally of finite presentation.

- (a') *$F \setminus G$ represents the (fppf) quotient sheaf $\widetilde{F \setminus G}$, and $G \rightarrow F \setminus G$ is an F -torsor locally trivial for the (fppf) topology.*
- (b) *$F \setminus G$ is flat over A , and is smooth over A if G is so.*
- (c) *$u : F \rightarrow G$ is a closed immersion, and $F \setminus G$ is separated.*
- (d) *If, moreover, F is a normal subgroup of G , there is a unique A -group structure on $F \setminus G$ for which $p : G \rightarrow F \setminus G$ is a morphism of A -groups.*

In the proof of this theorem, A' will denote an A -algebra which is local, finite, and free over A . If R is a relation involving A' , we shall say that “ $R(A')$ is true when A' is sufficiently large” if there is an A -algebra A_1 , local, finite, and free over A , such that the relation $R(A')$ holds for every A -algebra A' which is local, finite, and free over A_1 .

We shall first prove the theorem when F and G are of finite type over A .

3.2.1.— Suppose for a moment that every point of G has an open saturated neighborhood W such that the groupoid induced by G_* on W has a quasi-section (cf. Exposé V, § 6). Then, by Exposé V, 6.1, assertions (i), (ii), (iii), and (iv)(a) hold, and $F \setminus G$ is of finite type over A .^{*} Moreover, under the hypothesis of (iv), since

$$G \longrightarrow F \setminus G$$

is faithfully flat and locally of finite presentation, assertion (b) follows from EGA IV, 2.2.14 and 17.7.7. Assertion (iv)(a') follows from (iv)(a), by Exposé IV, 3.4.3.1, 5.2.2, and 5.1.6. Finally, assertion (iv)(c) will be proved in 3.2.5, while (ii') and (iv)(d) will be proved in Section 5.

We now prove the following assertion:

every finite subset of $F \setminus G$ is contained in an affine open subset. (†)

35

If U is a quasi-section of the groupoid induced by G_* on a saturated open subset W of G , then $U \otimes_A A'$ is a quasi-section of the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $W \otimes_A A'$. Moreover, if U_* is the (Sch/A) -groupoid induced by G_* on U , then $U_* \otimes_A A'$ is identified with the (Sch/A') -groupoid induced by $G_* \otimes_A A'$ on $U \otimes_A A'$.³⁶

It follows from the proofs in Exposé V that the construction of the quotient

$$X = F \setminus G$$

commutes with a base extension $A \rightarrow A'$ of the kind considered here.³⁷

^{*}Translator's note: the French source prints k here and again in editor's note 39. In this part of the proof the base is the Artinian local ring A , and “finite type over A ” is the required statement.

³⁵Editor's note: let us point out that if $A = k$ is a field, every quasi-compact open subset of $F \setminus G$ is quasi-projective (a result due to Chow for smooth algebraic groups); cf. [Ray70], VI, 2.6. By contrast, over the Artinian local ring $A = \mathbb{C}[\varepsilon]/(\varepsilon^2)$, there exist abelian A -schemes G which are not projective (loc. cit., XII, 4.2).

³⁶Editor's note: the preceding assertion is valid for every base change $A \rightarrow A'$.

³⁷Editor's note: this is explained in detail in 4.6 below. One must show that formation of direct images by the morphisms p , λ , and pr_2 commutes with flat base changes $A \rightarrow A'$. Since F and G are of finite type over the Artinian ring A , the morphisms f in question are all quasi-compact and quasi-separated, and

$$f_*(\mathcal{O}_X) \otimes_A A' = f'_*(\mathcal{O}_{X'})$$

(with the evident notation) follows from EGA IV₁, 1.7.21.

Let $x_1, \dots, x_n$ be points of $X = F \backslash G$, which we may suppose to be closed,³⁸ and let $g_1, \dots, g_n$ be closed points of G mapping to $x_1, \dots, x_n$. Let V be an everywhere dense affine open subset of X ,³⁹ and let U be its inverse image in G . By 1.2, there is an A -algebra A' , local, finite, and free over A , such that the points $g'_1, \dots, g'_p$ of

$$G' = G \otimes_A A'$$

lying above $g_1, \dots, g_n$ are strictly rational over A' .⁴⁰ Since the morphisms $G' \rightarrow G$ and $G \rightarrow X$ are open,

$$U' = U \otimes_A A'$$

is dense in G' . Hence the open subset

$$\bigcap_{i=1}^p (U')^{-1} \cdot g'_i$$

is nonempty and therefore contains a closed point x . By 1.2 (and 0.4.1), after enlarging A' if necessary, we may suppose that x is strictly rational over A' . Since $x \in (U')^{-1} \cdot g'_i$, we have

$$g'_i \in U' \cdot x \quad (1 \leq i \leq p).$$

Write V' for the inverse image of V in

$$X' = X \otimes_A A'.$$

It is an affine open subset of X' , and is also the image of U' under the projection $G' \rightarrow X'$. Since right translation r_x is an automorphism of the groupoid $G_* \otimes_A A'$, it induces an automorphism, again denoted r_x , of the quotient X' . Consequently,

$$V' \cdot x = r_x(V'),$$

the image of $U' \cdot x$ in X' , is an affine open subset of X' containing the images $x'_1, \dots, x'_p$ of $g'_1, \dots, g'_p$.

Consider the equivalence relation on $X' = X \otimes_A A'$ defined by the projection $X \otimes_A A' \rightarrow X$:

$$X \otimes_A A' \otimes_A A' \begin{array}{c} \xrightarrow{d_1} \\ \xrightarrow{d_0} \end{array} X \otimes_A A' \longrightarrow X, \quad ,$$

Here d_0 and d_1 are induced by the two canonical injections of A' into $A' \otimes_A A'$. Since A' is a finite free A -algebra, say of rank n , d_0 and d_1 are finite and locally free of rank n . We may therefore apply the argument of Exposé V, 5.b, taken from the proof of SGA 1, VIII.7.6. It

³⁸Editor's note: indeed, let $y_1, \dots, y_n$ be arbitrary points of X . Since X is of finite type over A , the closure of each y_i contains a closed point x_i , and every open subset containing x_i contains y_i .

³⁹Editor's note: such an open subset exists because X is of finite type over A : X has only finitely many irreducible components $C_1, \dots, C_p$, and it is enough, for each i , to choose a nonempty affine open subset contained in

$$C_i - \bigcup_{j \neq i} C_j.$$

(Here, moreover, one knows from (ii') that the C_i are disjoint.)

⁴⁰Editor's note: the original has been expanded in what follows.

follows that $x'_1, \dots, x'_p$ are contained in a saturated affine open subset W' contained in the affine open subset $V' \cdot x$. The image of W' in X then contains $x_1, \dots, x_n$, and is an affine open subset of X , by Exposé V, 4.1(ii).

3.2.2.— For every A -algebra A' , local, finite, and free over A , let $U(A')$ denote the set of points of $G \otimes_A A'$ having an open saturated neighborhood W such that the groupoid induced by $G_* \otimes_A A'$ on W has a quasi-section. It is clear that $U(A')$ is saturated under the operations of $G(\text{Spec } A')$ on $G \otimes_A A'$. We shall show that, when A' is sufficiently large,

$$U(A') = G \otimes_A A'.$$

By Theorem V.8.1, $U(A)$ is nonempty, and therefore contains a closed point y . We argue by induction on

$$\dim(G - U(A)).$$

Let $g_1, \dots, g_n$ be closed points belonging to the various irreducible components of $G - U(A)$. By 1.2, there is an A -algebra A' , local, finite, and free over A , such that the points

$$g'_1, \dots, g'_p \quad (\text{respectively, } x = x_1, \dots, x_r)$$

of $G' = G \otimes_A A'$, lying above $g_1, \dots, g_n$ (respectively, above y), are strictly rational over A' . Then $U(A')$ contains

$$(U(A) \otimes_A A') \cdot x^{-1}g'_i \quad \text{for every } i.$$

Consequently, $U(A')$ contains $g'_1, \dots, g'_p$, and

$$\dim(G' - U(A')) < \dim(G - U(A)).$$

The induction hypothesis therefore gives an algebra A'' , local, finite, and free over A' , such that

$$U(A'') = G' \otimes_{A'} A'' = G \otimes_A A''.$$

3.2.3.— We are now in a position to prove the existence of $F \backslash G$ when F and G are of finite type over A . Let A' be sufficiently large over A that $U(A')$ coincides with $G \otimes_A A'$ (cf. 3.2.2). Set

$$A'' = A' \otimes_A A',$$

and, for every A -scheme X , write X' and X'' for the fiber products $X \otimes_A A'$ and $X \otimes_A A''$. By 3.2.1 and 3.2.2, the quotients $F' \backslash G'$ and $F'' \backslash G''$ exist, and we have the following commutative diagram, in which the first two rows and columns are exact:

$$(*) \quad \begin{array}{ccccc} F'' \times_{A''} G'' & \xrightarrow{\text{pr}_2''} & G'' & \xrightarrow{p''} & F'' \backslash G'' \\ w_1 \downarrow & \lambda'' \searrow & v_1 \downarrow & & u_1 \downarrow \\ F' \times_{A'} G' & \xrightarrow{\text{pr}_2'} & G' & \xrightarrow{p'} & F' \backslash G' \\ h \downarrow & \lambda' \searrow & g \downarrow & & \\ F \times_A G & \xrightarrow{\text{pr}_2} & G & & \end{array}.$$

In this diagram, pr_2' and λ' (respectively, pr_2'' and λ'') are obtained from pr_2 and λ by the evident base changes; the morphisms g and h are induced by the canonical injection

$A \rightarrow A'$. We write p' and p'' for the canonical morphisms, while v_1, v_2 and w_1, w_2 are induced by the two canonical injections of A' into A'' .

Finally, formation of the quotient $F' \backslash G'$ commutes with the two base changes

$$f_1, f_2 : \operatorname{Spec} A'' \rightrightarrows \operatorname{Spec} A'.$$

Thus, if

$$\pi' : F' \backslash G' \rightarrow \operatorname{Spec} A'$$

denotes the structure morphism, there are canonical isomorphisms, for $i = 1, 2$,

$$\tau_i : F'' \backslash G'' \xrightarrow{\sim} (F' \backslash G') \times_{\pi', f_i} \operatorname{Spec} A'',$$

and u_i is the composite of τ_i with the projection

$$(F' \backslash G') \times_{\pi', f_i} \operatorname{Spec} A'' \rightarrow F' \backslash G'.$$

Now, for a diagram of type $(*)$ in which the first two rows and columns are exact, one checks readily that $\operatorname{Coker}(\operatorname{pr}_2, \lambda)$ exists if and only if $\operatorname{Coker}(u_1, u_2)$ exists, and that these two cokernels are then identified. The existence of $F \backslash G$ therefore follows from that of $\operatorname{Coker}(u_1, u_2)$.

The compatibility of the formation of $F \backslash G$ with the base extensions considered here (cf. editor's note 37 in 3.2.1 and 4.6 below) shows that the composite

$$(F' \backslash G') \times_{\pi', f_1} \operatorname{Spec} A'' \xrightarrow{\tau_1^{-1}} F'' \backslash G'' \xrightarrow{\tau_2} (F' \backslash G') \times_{\pi', f_2} \operatorname{Spec} A''$$

is a descent datum on $F' \backslash G'$ relative to

$$f : \operatorname{Spec} A' \rightarrow \operatorname{Spec} A.$$

By 3.2.1 (†) and SGA 1, VIII.7.6, this descent datum is effective; that is, $\operatorname{Coker}(u_1, u_2)$ exists. (One could instead apply Exposé V, Theorem 4.1 directly.)

3.2.4.— To complete the proof of assertions (i), (ii), (iii), and (iv)(a) of 3.2 in the case where F and G are of finite type over A , it remains to study the quotient $F \backslash G$. By Exposé V, 6.1, assertions (ii), (iii), and (iv)(a) “become true” after the base change

$$f : \operatorname{Spec} A' \rightarrow \operatorname{Spec} A.$$

By EGA IV₂, 2.6.1, 2.6.2, and 2.7.1, these assertions were therefore true before the base change. Finally, to prove the second assertion of (i), namely that $F \backslash G$ is the cokernel of $(\operatorname{pr}_2, \lambda)$ in the category of all ringed spaces, it suffices to refer to Exposé V, §6(c).

3.2.5.⁴¹ We now prove assertion (iv)(c) of 3.2, following the proof of Exposé VI_B, 9.2.1. Write

$$X = F \backslash G$$

and let d be the morphism

$$F \times_A G \rightarrow G \times_A G$$

whose components are λ and pr_2 .

Since u is a closed immersion by 2.5.2, and since

$$d = \sigma \circ (u \times \operatorname{id}_G), \quad \sigma(x, y) = (xy, y),$$

where σ is an automorphism of $G \times_A G$, the morphism d is a closed immersion. On the other hand, by (iv)(a), the following square is Cartesian:

⁴¹Editor's note: we have added this paragraph.

$$\begin{array}{ccc}
F \times_A G & \xrightarrow{d} & G \times_A G \\
\downarrow & & \downarrow p \times p \\
X & \xrightarrow{\Delta_X} & X \times_A X
\end{array}$$

Moreover, p , and hence also $p \times p$, is faithfully flat and locally of finite presentation. Therefore, by fppf descent, since d is a closed immersion, so is Δ_X ; that is, X is separated.

3.3.—⁴² In Theorem 3.2, the hypothesis that G be flat can be omitted when $u : F \rightarrow G$ is a monomorphism. This generalization is mentioned in Exposé VI_B, Remark 9.3(b), and also in [Ray67a], Example a)(i), p. 82. The proof, given in [An73], Theorem 4, follows from Theorem 3.2 and from the following theorem of Grothendieck (mentioned in [Ray67a], Theorem 1(ii), and proved in [DG70], § III.2, 7.1). If X is a scheme and R an equivalence relation on X , we write $\widetilde{X/R}$ for the (fppf) quotient sheaf of X by R (cf. Exposé IV, 4.4.9).

Theorem 3.3.1 (Grothendieck). *Let A be a ring, let X be an A -scheme, and let*

$$d_0, d_1 : R \rightrightarrows X$$

be an A -equivalence relation on X , with each d_i faithfully flat and of finite presentation. Let X_0 be a saturated closed subscheme of X , defined by a nilpotent ideal, and let R_0 be the equivalence relation induced by R on X_0 . If the (fppf) quotient sheaf $\widetilde{X_0/R_0}$ is representable by an A -scheme, then so is $\widetilde{X/R}$.

For the proof, see [DG70], § III.2, 7.1. Let us now return to the case where A is an Artinian local ring. Let $u : F \rightarrow G$ be a quasi-compact morphism between A -groups locally of finite type, and suppose in addition that u is a monomorphism. Then u is a closed immersion by 2.5.2.

We may now state the following variant of Theorem 3.2.

Theorem 3.3.2. *Let A be an Artinian local ring, let G be an A -group locally of finite type, and let F be a closed subgroup of G , flat over A .⁴³ Then:*

- (i) *The (fppf) quotient sheaf $\widetilde{F \backslash G}$ is representable by a separated A -scheme $F \backslash G$, locally of finite type; moreover, the sequence*

$$F \times_A G \rightrightarrows G \xrightarrow{p} F \backslash G$$

is exact in the category of all ringed spaces.

- (ii) *The morphism*

$$F \times_A G \xrightarrow{(\lambda, \text{pr}_2)} G \times_{(F \backslash G)} G$$

is an isomorphism, and $p : G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation; hence p is an F -torsor locally trivial for the (fppf) topology.

- (iii) *If G is flat (respectively, of finite type; respectively, smooth) over A , then so is $F \backslash G$.*
- (iv) *$X = F \backslash G$ is equipped with a right action of G such that $p(e) \cdot g = p(g)$ for every $g \in G$. Consequently, the connected components of X are of finite type over A , irreducible, and all of dimension $\dim G - \dim F$.*

⁴²Editor's note: we have added this subsection.

⁴³Editor's note: for an example in which F is not flat and $\widetilde{F \backslash G}$ is not representable, see [DG70], § III.3, no. 3.3.

- (v) If, in addition, F is a normal subgroup of G , there is a unique A -group structure on $F \backslash G$ for which $p : G \rightarrow F \backslash G$ is a morphism of A -groups.

Assertions (i) and (ii) follow from 3.2 and 3.3.1. Since $G \rightarrow F \backslash G$ is faithfully flat and locally of finite presentation, assertion (iii) follows from EGA IV, 2.2.14, 2.7.1, and 17.7.7. Assertions (iv) and (v) will be proved in Section 5. Let us record at once the following corollary.

Corollary 3.3.3. *Let A be an Artinian local ring, let G be an A -group locally of finite type, and let H be a closed subgroup of G , flat over A . Write p for the morphism $G \rightarrow G/H$, and write λ (respectively, pr_1) for the morphism $G \times H \rightarrow G$ defined by $\lambda(g, h) = gh$ (respectively, the projection $G \times H \rightarrow G$). Then, for every open subset U of G/H ,*

$$\mathcal{O}(U) = \{\varphi \in \mathcal{O}(p^{-1}(U)) \mid \varphi \circ \lambda = \varphi \circ \text{pr}_1\}.$$

In other words, $\mathcal{O}(U)$ is the set of $\varphi \in \mathcal{O}(p^{-1}(U))$ such that $\varphi(gh) = \varphi(g)$ for every A -scheme S and every $g \in G(S)$, $h \in H(S)$.

Indeed, $p : G \rightarrow G/H$ is faithfully flat and locally of finite presentation, hence a covering for the (fppf) topology; the result follows from Exposé IV, 3.3.3.2.

4. Construction of quotients $F \backslash G$ (general case)

We now assume the hypotheses of Theorem 3.2, without requiring F and G to be of finite type over A .

4.1.— Let G^α be a connected component of G . We first show that the saturation

$$\mathcal{S}(G^\alpha)$$

⁴⁴ of G^α under the equivalence relation defined by the groupoid G_* is an open-and-closed subset of G ; in other words, it is a union of connected components of G .

This saturation is the image of $F \times_A G^\alpha$ under λ , and is therefore open in G (cf. § 3.1). Let k be the residue field of A , and let $\bar{k}$ be an algebraic closure of k . It remains to show that the image of

$$(F \times_A G^\alpha) \otimes_A k$$

under $\lambda \otimes_A k$ is closed in $G \otimes_A k$, or, equivalently by SGA 1, VIII.4.4, that the image of

$$(F \times_A G^\alpha) \otimes_A \bar{k}$$

under $\lambda \otimes_A \bar{k}$ is closed in $G \otimes_A \bar{k}$. Since $G^\alpha \otimes_A \bar{k}$ is a union of finitely many connected components of $G \otimes_A \bar{k}$, we are reduced to the case where A is an algebraically closed field, which we now assume. In this case, $\mathcal{S}(G^\alpha)$ is the union of the images of G^α under the left translations $\ell_{u(x)}$, where x runs through the closed points of F . The assertion follows because these images are connected components of G .

4.2.— Take G^α in particular to be the connected component G^0 of the identity in G . Then $\mathcal{S}(G^0)$ plainly contains the image of F under u , which is precisely the equivalence class of the identity. On the other hand, if F^β is a connected component of F , then $F^\beta \times_A G^0$ is connected by 2.1.2, so its image under λ is contained in the connected component of $u(F^\beta)$ in G . In other words, $\mathcal{S}(G^0)$ is the union of the connected components that meet the image of F .

⁴⁴Editor's note: we have written $\mathcal{S}(G^\alpha)$, rather than $\overline{G^\alpha}$, for the saturation of G^α .

Let us also observe that the open subscheme of G whose underlying space is $\mathcal{S}(G^0)$ is a subgroup of G ; we continue to denote it by $\mathcal{S}(G^0)$. Indeed, inversion on G preserves the image of F and permutes the connected components of G that meet this image. It therefore suffices to prove that

$$\nu : G \times_A G \longrightarrow G$$

maps $\mathcal{S}(G^0) \times_A \mathcal{S}(G^0)$ into $\mathcal{S}(G^0)$. For this we may suppose that A is an algebraically closed field: with the notation of 4.1, $\mathcal{S}(G^0) \otimes_A \bar{k}$ is the saturation of $(G \otimes_A \bar{k})^0$ for the equivalence relation defined by the homomorphism $u \otimes_A \bar{k}$.

If G^γ and G^δ are connected components of $\mathcal{S}(G^0)$, then $G^\gamma \times_A G^\delta$ is connected and its image under ν meets the image of F . Consequently,

$$\nu(G^\gamma \times_A G^\delta)$$

is contained in a connected component of G that meets $u(F)$.*

4.3.— It follows from the preceding discussion that the groupoid G_* , with base G , defined by u , is the direct sum of the groupoids $\mathcal{S}(G_*^\alpha)$ induced by G_* on the various open-and-closed parts of G of the form $\mathcal{S}(G^\alpha)$. The cokernel of G_* is therefore the direct sum of the cokernels of the groupoids $\mathcal{S}(G_*^\alpha)$, which we are thus led to study separately.

First consider the groupoid $\mathcal{S}(G_*^0)$ induced by G_* on $\mathcal{S}(G^0)$. It is clear that $\mathcal{S}(G_*^0)$ is the groupoid with base $\mathcal{S}(G^0)$ defined by the homomorphism

$$F \longrightarrow \mathcal{S}(G^0)$$

induced by u (§ 3.1). The cokernel whose existence we wish to prove is thus identified with $F \backslash \mathcal{S}(G^0)$. Consider, on the other hand, the groupoid

$$\begin{array}{ccccc} & \xrightarrow{\ell'_2} & & \xrightarrow{\ell_1} & \\ G_2^0 & \xrightarrow{\ell'_1} & G_1^0 & \xrightarrow{\ell_0} & G_0^0 = G^0 \\ & \xrightarrow{\ell'_0} & & & \end{array}$$

induced by $\mathcal{S}(G_*^0)$ on G^0 . Referring to the explicit construction in Exposé V, § 3.b, the object denoted there by $Y_0 \times_{X_0} X_1$ is precisely $F \times_A G^0$. Hence G_1^0 is the inverse image of G^0 under the morphism

$$F \times_A G^0 \longrightarrow \mathcal{S}(G^0)$$

induced by λ .

I claim that this inverse image is $F_0 \times_A G^0$, where F_0 denotes the inverse image of G^0 under u . Indeed, if F^β is a connected component of F_0 , then $F^\beta \times_A G^0$ is connected by 2.1.2 and $\lambda(F^\beta \times_A G^0)$ is contained in G^0 .* Conversely, if F^β is a connected component of F not contained in F_0 , then the image of $F^\beta \times_A G^0$ is again connected and contains $u(F^\beta)$; if $u(F^\beta)$ is not contained in G^0 , then $\lambda(F^\beta \times_A G^0)$ does not meet G^0 .

It follows that the groupoid G_*^0 induced by G_* on G^0 is the groupoid with base G^0 defined by the homomorphism

$$F_0 \longrightarrow G^0$$

*Translator's note: the French source prints $u(G^\gamma \times_A G^\delta)$ in this final sentence. The map defined and used throughout the paragraph is the multiplication $\nu : G \times_A G \rightarrow G$; the target therefore uses ν .

*Translator's note: the French source prints $\lambda(F \times_A G^0)$ at this point. The statement concerns the connected component F^β just introduced, and the converse immediately below confirms the intended restriction to $F^\beta \times_A G^0$.

induced by u . Since G^0 , and hence F_0 , are of finite type over A , Section 3 shows that G_*^0 has a cokernel, namely $F_0 \backslash G^0$.[†]

I now claim that $F_0 \backslash G^0$ is identified with $F \backslash \mathcal{S}(G^0)$. The proof is analogous to that of the first part of assertion (i) of Exposé V, Lemma 6.1. Consider the diagram

$$\mathcal{S}(G^0) \xleftarrow{v} F \times_A G^0 \xrightarrow{\text{pr}_2} G^0,$$

where v is the morphism induced by λ . Since pr_2 has a section, it is a universal effective epimorphism. Consequently, $F_0 \backslash G^0$ coincides with $\text{Coker}(v_0, v_1)$, where

$$\begin{array}{ccccc} & \xrightarrow{v'_2} & & & \\ V_2 & \xrightarrow{v'_1} & V_1 & \xrightarrow[v_0]{v_1} & V = F \times_A G^0 \\ & \xrightarrow{v'_0} & & & \end{array}$$

is the inverse image under pr_2 of the groupoid G_*^0 (cf. Exposé V, § 3.a), and hence also the inverse image of $\mathcal{S}(G_*^0)$ under the composite morphism

$$F \times_A G^0 \xrightarrow{\text{inclusion}} F \times_A \mathcal{S}(G^0) \xrightarrow{\text{pr}_2} \mathcal{S}(G^0).$$

Similarly, since v is faithfully flat and quasi-compact, $F \backslash \mathcal{S}(G^0)$ coincides with the cokernel of the inverse image of $\mathcal{S}(G_*^0)$ under the base change v . This inverse image is isomorphic to V_* by Exposé V, § 3.c. It follows that the canonical inclusion of G_*^0 in $\mathcal{S}(G_*^0)$ induces an isomorphism

$$F_0 \backslash G^0 \xrightarrow{\sim} F \backslash \mathcal{S}(G^0).$$

Finally, formation of $F \backslash \mathcal{S}(G^0)$ commutes with finite locally free base changes, because the same holds for $F_0 \backslash G^0$ (cf. editor's note 37 and 4.6 below).

4.4.— It remains to construct the cokernel of the groupoid $\mathcal{S}(G_*^\alpha)$ when G^α is an arbitrary connected component of G . Let A' be a sufficiently large local A -algebra, finite and free over A (cf. 3.2). Then $G^\alpha \otimes_A A'$ is the union of finitely many connected components

$$C^1, \dots, C^n$$

of $G \otimes_A A'$, each of which has a strictly rational point. For every i , there is therefore a right translation r_i of $G \otimes_A A'$ which maps $G^0 \otimes_A A'$ onto C^i . This translation induces an isomorphism

$$\mathcal{S}(G_*^0) \otimes_A A' \xrightarrow{\sim} \mathcal{S}(C_*^i).$$

Consequently, the groupoid induced by $G_* \otimes_A A'$ on the saturation of C^i has a cokernel.

Since $\mathcal{S}(G_*^\alpha) \otimes_A A'$ is the direct sum of some of the groupoids $\mathcal{S}(C_*^i)$, it too has a cokernel. This cokernel is a direct sum of some number of copies of

$$(F_0 \otimes_A A') \backslash (G^0 \otimes_A A').$$

Thus every finite subset of it is contained in an affine open subset. Moreover, formation of this cokernel commutes with finite locally free extensions of the base (cf. editor's note 37 and 4.6 below). As in 3.2.3, it follows that this cokernel has the form $Y \otimes_A A'$, where Y is a cokernel of $\mathcal{S}(G_*^\alpha)$.

[†]Translator's note: the French source says "by paragraph 4." The finite-type quotient invoked here is the result of Section 3; Section 4 is the general-case construction currently in progress.

4.5.— We have now constructed $F \backslash G$ and shown that it is a direct sum of schemes of finite type over A . The remaining assertions of Theorem 3.2 reduce directly to assertions about the groupoids $\mathcal{S}(G_*^\alpha)$. As in Exposé V, § 6, the second assertion in (i) follows from the first together with (ii) and (iii); it is therefore enough to prove (ii), (iii), and (iv)(a).

Since A' is a local A -algebra, finite and free over A , the morphism $A \rightarrow A'$ is faithfully flat and of finite presentation. By SGA 1, VIII (3.1, 4.6, 5.4), it is therefore enough to verify the corresponding assertions for the groupoid

$$\mathcal{S}(G_*^\alpha) \otimes_A A'.$$

This groupoid is isomorphic to a direct sum of finitely many copies of $\mathcal{S}(G_*^0) \otimes_A A'$ (cf. 4.4), so we are reduced to the groupoid $\mathcal{S}(G_*^0)$. For this last groupoid we continue to follow the proof given in Exposé V, § 6, as we began to do in 4.3.

4.6.— Let us conclude this section with some remarks concerning Lemma 6.1 and § 9(a) of Exposé V. Under the hypotheses and with the notation of Exposé V, § 9(a), we seek a condition under which formation of the cokernel of the (Sch/S) -groupoid X_* commutes with a base extension

$$\pi : S' \longrightarrow S.$$

The cokernels of X_* and X'_* are identified with the cokernels of the groupoids U_* and U'_* induced by X_* and X'_* on the quasi-sections U and U' , respectively. We are therefore reduced to the case of a (Sch/S) -groupoid satisfying the hypotheses of Exposé V, Theorem 4.1.

⁴⁵Let Y be the cokernel of U_* , put

$$Y' = Y \times_S S',$$

and let Y_1 be the cokernel of U'_* ; we saw in Exposé V, § 9(a), that the canonical morphism

$$Y_1 \longrightarrow Y'$$

is a homeomorphism (indeed, a universal homeomorphism); we may thus identify Y_1 and Y' as topological spaces. If $p : U \rightarrow Y$ is the canonical morphism and $p' : U' \rightarrow Y'$ is obtained from it by base change, we want the following sequence of $\mathcal{O}_{Y'}$ -modules to be exact:

$$\mathcal{O}_{Y'} \longrightarrow p'_*(\mathcal{O}_{U'}) \rightrightarrows p'_*u'_{1*}(\mathcal{O}_{U'_1}) = p'_*u'_{0*}(\mathcal{O}_{U'_1}). \quad (*)$$

⁴⁶Since we are under the hypotheses of Exposé V, Theorem 4.1, u_0 and u_1 are finite and locally free; moreover, p is integral by Exposé V, 4.1(ii). Thus p and $p \circ u_i$ are affine, hence separated and quasi-compact.

Consequently, if S' is flat over S , EGA III₁, 1.4.15 (taking account of correction Err_{III} 25 in EGA III₂) identifies the sequence (*) with the inverse image of the sequence

$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_U) \rightrightarrows p_*u_{1*}(\mathcal{O}_{U_1}) = p_*u_{0*}(\mathcal{O}_{U_1}), \quad (**)$$

which is exact.⁴⁷ An analogous argument applies when the groupoid X_* has quasi-sections “locally” (cf. the proof of Exposé V, Theorem 7.1). We therefore obtain:

⁴⁵Editor’s note: we have added the following sentence.

⁴⁶Editor’s note: in what follows we have modified the original; the additional hypotheses imposed there on X_* are unnecessary.

⁴⁷Editor’s note: we have added both the following sentence and the numbering 4.6.1 below, in order to highlight the stated result.

Proposition 4.6.1. *Formation of the cokernel of X_* commutes with flat base extensions when X_* has quasi-sections locally.*

4.7.— Consider the groupoid G_* of Theorem 3.2, supposing temporarily that F and G are of finite type over A . By 3.2.2 there is a local A -algebra A' , finite and free over A , such that $G_* \otimes_A A'$ has quasi-sections “locally.” For every base extension $T \rightarrow \operatorname{Spec} A$, the sequence

$$(F'' \backslash G'') \times_{\operatorname{Spec} A} T \rightrightarrows (F' \backslash G') \times_{\operatorname{Spec} A} T \longrightarrow (F \backslash G) \times_{\operatorname{Spec} A} T$$

obtained from diagram (*) of 3.2.3 is exact.

If, in addition, T is flat over $\operatorname{Spec} A$, then 4.6 identifies

$$(F'' \backslash G'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (F' \backslash G') \times_{\operatorname{Spec} A} T$$

respectively with the cokernels of the groupoids

$$(G_* \otimes_A A'') \times_{\operatorname{Spec} A} T \quad \text{and} \quad (G_* \otimes_A A') \times_{\operatorname{Spec} A} T.$$

The diagram obtained from 3.2.3(*) by the base change $T \rightarrow \operatorname{Spec} A$ then shows that

$$(F \backslash G) \times_{\operatorname{Spec} A} T$$

is the cokernel of $G_* \times_{\operatorname{Spec} A} T$. The same argument applies in the general case, when F and G are merely locally of finite type over A . Thus:

Proposition 4.7.1. *Under the hypotheses of Theorem 3.2, for every flat A -scheme T , the scheme*

$$(F \backslash G) \times_{\operatorname{Spec} A} T$$

is identified with the left quotient of $G \times_{\operatorname{Spec} A} T$ by $F \times_{\operatorname{Spec} A} T$.⁴⁸

5. Connections with Exposé IV and consequences

5.1.—⁴⁹ We retain the notation of § 3 and the hypotheses of Theorem 3.2. We then have the following commutative diagram:

$$\begin{array}{ccc} F \times_A G \times_A G & \xrightarrow{F \times \nu} & F \times_A G \\ \text{pr}_2 \times G \downarrow \lambda \times G & & \text{pr}_2 \downarrow \lambda \\ G \times_A G & \xrightarrow{\nu} & G \\ p \times G \downarrow & & \downarrow p \\ (F \backslash G) \times_A G & \xrightarrow{\rho} & F \backslash G \end{array},$$

It satisfies

$$\text{pr}_2 \circ (F \times \nu) = \nu \circ (\text{pr}_2 \times G), \quad \lambda \circ (F \times \nu) = \nu \circ (\lambda \times G).$$

Moreover, since G is flat over A , the left-hand vertical sequence is exact by 4.7. Hence ν induces a morphism of A -schemes

$$\rho : (F \backslash G) \times_A G \longrightarrow F \backslash G.$$

⁴⁸Editor's note: we have added the numbering 4.7.1 in order to highlight this result.

⁴⁹Editor's note: we have changed the title of this section, which was called “Complements” in the original.

The morphism ρ makes G act on the right on $F \backslash G$, as is immediately checked; moreover, the canonical morphism $G \rightarrow F \backslash G$ commutes with the right actions of G on G and on $F \backslash G$.⁵⁰ This proves the first assertion of Theorem 3.2(ii'). By 2.5.4, the connected components of

$$X = F \backslash G$$

are therefore of finite type, irreducible, and all of the same dimension.

To compute this dimension, we may suppose that $A = k$ and that k is algebraically closed. By Exposé I, 2.3.3.1, the stabilizer of the k -point $p(e)$ is represented by the fiber

$$H = p^{-1}(p(e)).$$

Since $F \backslash G$ is the quotient of G by F in the category of ringed spaces, the underlying space of this fiber is $u(F)$. As $\ker(u)$ is finite, we obtain

$$\dim H = \dim u(F) = \dim F.$$

It follows from 2.5.4(ii) that

$$\dim X = \dim G - \dim F.$$

This proves Theorem 3.2(ii'), and hence also Theorem 3.3.2(iv).

5.2.— When the homomorphism of A -groups $u : F \rightarrow G$ is a monomorphism, the result of 5.1 may also be recovered from the results of Exposé IV. Indeed, the canonical morphism

$$p : G \longrightarrow F \backslash G$$

is faithfully flat and open by 3.2; it is therefore a covering for the (fpqc) topology by Exposé IV, 6.3.1, and we may apply Exposé IV, Corollaries 5.2.2 and 5.2.4.

In particular, suppose, in addition to the hypotheses of 3.2, that u is the inclusion in G of a normal subgroup F . Then there is a unique A -group structure on $F \backslash G$ for which the canonical morphism

$$p : G \longrightarrow F \backslash G$$

is a homomorphism of A -groups.⁵¹ This proves Theorem 3.3.2(v).

5.3.— We now review several statements from Exposé IV.

5.3.1.— The statements of Exposé IV, 5.2.7 and 5.3.1 translate as follows. Let F and G be groups locally of finite type and flat over A , with F a closed normal subgroup of G . The maps

$$H \mapsto F \backslash H, \quad H' \mapsto H' \times_{(F \backslash G)} G$$

define a bijection between the flat A -subgroups of G containing F and the flat A -subgroups of $F \backslash G$. Under this bijection, the closed (respectively, normal) subgroups of G containing F correspond to the closed (respectively, normal) subgroups of $F \backslash G$.⁵²

5.3.2.— Exposé IV, Proposition 5.2.9, gives the following result. Let F, H, G be groups locally of finite type and flat over A , and suppose

$$F \subset H \subset G,$$

⁵⁰Editor's note: we have added what follows.

⁵¹Editor's note: we have added the following sentence.

⁵²Editor's note: in addition to the cited statements of Exposé IV, we use the fact that, since $G \rightarrow F \backslash G$ is faithfully flat, an A -subgroup H of G is flat over A if and only if $F \backslash H$ is.

with F closed in G and normal in H . Then $F \backslash H$ acts freely on the left on $F \backslash G$, the quotient scheme

$$(F \backslash H) \backslash (F \backslash G)$$

exists, and there is a canonical isomorphism of G -schemes

$$(F \backslash H) \backslash (F \backslash G) \xrightarrow{\sim} H \backslash G.$$

5.3.3.— Finally, Exposé IV, 5.2.8, gives the following assertion. Let F, H, G be groups locally of finite type and flat over A . Suppose that F is contained in G , closed and normal in G , that H is contained in G , and that $F \cap H$ is flat over A . Let

$$F \times_A^\tau H$$

be the A -group whose underlying scheme is $F \times_A H$, with multiplication defined by

$$((x, h), (y, h')) \mapsto (xhyh^{-1}, hh').$$

Likewise, let

$$u : H \cap F \longrightarrow F \times_A^\tau H$$

be the monomorphism $x \mapsto (x^{-1}, x)$, and let $F \cdot H$ be the quotient

$$(F \cap H) \backslash (F \times_A^\tau H).$$

Then there is a canonical isomorphism

$$F \backslash (F \cdot H) \xrightarrow{\sim} (F \cap H) \backslash H.$$

5.4.—⁵³ Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, such that the kernel N of u is flat over A . In this case, by 3.3.2 and 5.2, the quotient A -group

$$C = N \backslash G$$

exists, and the morphism $p : G \rightarrow C$ is faithfully flat and locally of finite presentation. On the other hand, by Exposé IV, 5.2.6, u induces a monomorphism

$$v : C \longrightarrow H,$$

which is quasi-compact (because u is quasi-compact and $G \rightarrow C$ is surjective; cf. EGA IV₁, 1.1.3), and is therefore a closed immersion by 2.5.2. We have thus obtained the following proposition.

Proposition 5.4.1.— *Let $u : G \rightarrow H$ be a quasi-compact morphism between A -groups locally of finite type, such that $N = \ker u$ is flat over A . Then u admits the factorization*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ N \backslash G & & \end{array}$$

⁵³Editor's note: we have expanded the original in what follows; in particular, we have added Proposition 5.4.1.

where p is faithfully flat and locally of finite presentation, and i is a closed immersion.

Suppose in addition that G is flat over A . Then, by 3.3.2, $C = N \backslash G$ is flat over A , and hence the quotient

$$X = C \backslash H$$

exists in $(\mathbf{Sch}/_A)$ and represents the quotient (fppf)-sheaf $\widetilde{C} \backslash H$, while

$$q : H \longrightarrow X$$

is a C -torsor. Consequently, if

$$e : \mathrm{Spec} A \longrightarrow H$$

denotes the unit section of H , then v induces an isomorphism of (fppf)-sheaves between $\widetilde{C}$ and the fiber product of q with

$$q \circ e : \mathrm{Spec} A \longrightarrow X,$$

which is represented by a closed subscheme of H . Consequently, v is an isomorphism of C onto a closed group subscheme K of H , equal to the stabilizer of the A -point $q \circ e$ of X .^{*} (This also gives another proof that every quasi-compact monomorphism $v : C \rightarrow H$ between A -groups locally of finite type is a closed immersion; cf. 2.5.2 and Exposé VI_B, 1.4.2.)

Suppose further that C is a normal subgroup of H . In this case, the A -group

$$\overline{H} = C \backslash H$$

is the cokernel, in the category of A -groups, of the morphism $u : G \rightarrow H$, and K is the kernel of the morphism

$$H \longrightarrow \overline{H}.$$

When G and H are abelian A -groups, K is the image of u in the category of abelian A -groups, whereas

$$C = (\ker u) \backslash G$$

is the coimage of u . Taking into account the isomorphism

$$C \xrightarrow{\sim} K$$

just established, we obtain the following.⁵³

Theorem 5.4.2.— *Let k be a field. The category of commutative algebraic k -groups is abelian.*

Indeed, when k is a field, $\ker(u)$ is flat over k , whatever u may be.

⁵⁴ Note that the full subcategory of commutative affine algebraic k -groups is thick. Indeed, consider an exact sequence of commutative algebraic k -groups

$$1 \longrightarrow N \longrightarrow G \longrightarrow G/N \longrightarrow 1.$$

If G is affine, then N is clearly affine, and G/N is also affine by a theorem of Chevalley; cf. Exposé VI_B, 11.17. Conversely, if N and G/N are affine, then G is also affine by Exposé VI_B, 9.2(viii). We therefore obtain the following corollary.

^{*}Translator's note: the French source declares $e : \mathrm{Spec} A \rightarrow G$, although the displayed composition $q \circ e$ requires the unit section of H , and then prints "a closed group subscheme K of G ." Here $v : C \rightarrow H$, the displayed fiber product is represented inside H , and the next paragraph identifies K with $\ker(H \rightarrow \overline{H})$; accordingly, e is taken to be the unit section of H , and the intended ambient group of K is H .

⁵³Editor's note: we have added the number 5.4.2 to this theorem.

⁵⁴Editor's note: we have added what follows.

Corollary 5.4.3.— *Let k be a field. The category of commutative affine algebraic k -groups is abelian.*

Let us also point out that the category of all commutative affine k -groups (not necessarily of finite type) is abelian. This follows from Exposé VI_B, 11.17 and 11.18.2 (cf. [DG70], § III.3, 7.4); see also Exposé VII_B, 2.4.2 for a proof using formal groups.

5.5.— Let G be a group locally of finite type and flat over an Artinian local ring A . We know from 2.3 that the connected component G^0 of the identity is a normal open group subscheme of G , hence is also flat over A . It then follows from 3.2 and 5.2 that

$$G^0 \backslash G$$

is an A -group scheme, flat over A . Moreover, since every connected component G^α of G is saturated for the equivalence relation defined by G^0 , the quotient $G^0 \backslash G$ is the disjoint union of the $G^0 \backslash G^\alpha$ (cf. 4.3). In particular, the connected component of the identity in $G^0 \backslash G$ is precisely

$$G^0 \backslash G^0 \simeq \operatorname{Spec} A,$$

and hence the morphism

$$G^0 \backslash G \longrightarrow \operatorname{Spec} A$$

is a local isomorphism at the identity. Consequently, $G^0 \backslash G$ is étale over $\operatorname{Spec} A$, by Exposé VI_B, 1.3.⁵⁵ We therefore obtain the following proposition (for assertion (ii), see [DG70], § II.5, 1.7–1.10).

Proposition 5.5.1.— *Let A be an Artinian local ring and G an A -group locally of finite type and flat.*

- (i) $G^0 \backslash G$ is an étale A -group.
- (ii) *Consequently, if $A = k$ is an algebraically closed field, $G^0 \backslash G$ is a constant k -group acting simply transitively on the set of connected components of G . Thus, if G is algebraic, $G^0 \backslash G$ is finite.*

5.6.— Now let k be a perfect field and G a k -group locally of finite type. We saw in 0.2 that G_{red} is then a group subscheme of G . Moreover, the equivalence class of the identity of G under the left action of G_{red} on G is the whole underlying space of G . Theorem 3.2 therefore gives the following result.

Proposition 5.6.1.— *Let k be a perfect field and G a k -group locally of finite type. Then the k -scheme*

$$G_{\text{red}} \backslash G$$

is the spectrum of a finite local k -algebra with residue field k .

⁵⁶ Indeed, by 3.2, the quotient $G_{\text{red}} \backslash G$ has a single point, with residue field k , and is a k -scheme of finite type. It is therefore the spectrum of a finite-dimensional local k -algebra (cf. EGA I, 6.4.4).

Proposition 5.6.2.— *Let $u : F \rightarrow G$ be a morphism between groups locally of finite type over a perfect field k . The following assertions are equivalent:*

- (i) u is flat.

⁵⁵Editor's note: we have expanded the preceding discussion from the original and added Proposition 5.5.1 in order to make this result explicit.

⁵⁶Editor's note: we have added the number 5.6.1, as well as the proof that follows.

(ii) *The morphism*

$$u^0 : F^0 \longrightarrow G^0$$

is dominant, and the morphism

$$v : F_{\text{red}} \setminus F \longrightarrow G_{\text{red}} \setminus G$$

induced by u is flat.

⁵⁷ Indeed, consider the following commutative diagram:

$$\begin{array}{ccc} F & \xrightarrow{p} & F_{\text{red}} \setminus F \\ u \downarrow & & \downarrow v \\ G & \xrightarrow{q} & G_{\text{red}} \setminus G \end{array} \quad ,$$

where p and q denote the canonical projections. By 3.2(iv), p and q are faithfully flat. Consequently, if u is flat, then

$$q \circ u = v \circ p$$

is flat, and hence v is flat as well.

Conversely, suppose that v is flat and u^0 is dominant. Since u^0 is quasi-compact (the scheme F^0 is of finite type over k by 2.4, and hence Noetherian), it maps the generic point ξ of F^0 to the generic point η of G^0 . Let R be the finite local k -algebra such that

$$G_{\text{red}} \setminus G = \text{Spec } R,$$

and let $\mathfrak{m}$ be its maximal ideal. We have local homomorphisms of local rings

$$R \longrightarrow \mathcal{O}_{G,\eta} \longrightarrow \mathcal{O}_{F,\xi}.$$

Observe that the following square is Cartesian:

$$\begin{array}{ccc} G_{\text{red}} & \longrightarrow & G \\ \downarrow & & \downarrow q \\ \text{Spec}(R/\mathfrak{m}) & \longrightarrow & \text{Spec}(R) \end{array}$$

It follows that

$$\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta} \simeq \mathcal{O}_{G_{\text{red}},\eta} = \kappa(\eta),$$

so that

$$\mathcal{O}_{F,\xi}/\mathfrak{m}\mathcal{O}_{F,\xi}$$

is flat over $\mathcal{O}_{G,\eta}/\mathfrak{m}\mathcal{O}_{G,\eta}$.

On the other hand, since q and $v \circ p$ are flat, G and F are flat over R . The local criterion for flatness (cf. EGA IV₃, 11.3.10.2) therefore shows that $\mathcal{O}_{F,\xi}$ is flat over $\mathcal{O}_{G,\eta}$; in other words, u is flat at ξ . It follows from 2.5.3 that u is flat.

⁵⁷Editor's note: we have expanded the proof of (ii) $\Rightarrow$ (i) and simplified the diagram below.

6. Complements on k -groups not necessarily of finite type

⁵⁸ Let us also record the following results, which will be useful in the addendum, Exposé VI_B, § 12. We fix a base field k .

Lemma 6.1.— *Let G be a k -group. For every $x \in G$, there exists a point $u \in G \times G$ such that $\mu(u) = x$ and both projections $p_1(u)$ and $p_2(u)$ are maximal points of G .*

Proof. Set $K = \kappa(x)$. Since the projection $G_K \rightarrow G$ sends maximal points to maximal points, we are reduced to the case in which x is rational. Left translation λ_x (respectively, right translation ρ_x) then gives a morphism

$$G \longrightarrow G \times G, \quad g \longmapsto (\lambda_x(g^{-1}), g)$$

(respectively, $g \longmapsto (g, \rho_x(g^{-1}))$), which induces an isomorphism from G onto $\mu^{-1}(x)$, inverse to p_2 (respectively, p_1). Thus, if u is a maximal point of $\mu^{-1}(x)$, then $p_1(u)$ and $p_2(u)$ are maximal points of G , and u has the required properties.

Corollary 6.2.— *Let $f : G \rightarrow H$ be a quasi-compact dominant morphism of k -groups.*

(i) *f is surjective.*

(ii) *If H is reduced, f is faithfully flat.*

Proof. Write μ_H (respectively, μ_G) for the multiplication of H (respectively, G). Let $h \in H$. By 6.1, there exists $u \in H \times H$ such that $\mu_H(u) = h$ and $\alpha = p_1(u)$, $\beta = p_2(u)$ are maximal points of H . Since f is quasi-compact and dominant, $f^{-1}(\alpha)$ and $f^{-1}(\beta)$ are nonempty (cf. EGA IV₁, 1.1.5). Hence there exists $v \in G \times G$ such that

$$(f \times f)(v) = u$$

(cf. EGA I, 3.5.2). Then $g = \mu_G(v)$ satisfies $f(g) = h$. This proves that f is surjective.

Suppose in addition that H is reduced. Then $\mathcal{O}_{H,\alpha}$ is a field, and we have seen above that $f^{-1}(\alpha) \neq \emptyset$. Thus f is flat at every point $\xi \in f^{-1}(\alpha)$, and consequently f is flat by Lemma 2.5.3.

Reminder 6.3.— Recall (cf. EGA IV₃, 11.10.1) that a morphism of schemes $f : X \rightarrow Y$ is said to be *schematically dominant* if it satisfies the following condition: for every open subset U of Y , if Z is a closed subscheme of U such that the morphism $f^{-1}(U) \rightarrow U$ factors through Z , then $Z = U$. When f is quasi-compact and quasi-separated, this is equivalent to saying that the closed image of X under f is Y (cf. loc. cit., 11.10.3(iv), and EGA I, 9.5.8).

Proposition 6.4.— *Let $f : H \rightarrow G$ be a quasi-compact morphism of k -groups. Then the closed image of f is a group subscheme H' of G , and f factors as*

$$\begin{array}{ccc} H & \xrightarrow{f} & G \\ f' \downarrow & \nearrow i & \\ H' & & \end{array}$$

where f' is schematically dominant, quasi-compact, and surjective.

Proof. Since H is separated (0.3), f is quasi-compact and separated. Thus $f_*(\mathcal{O}_H)$ is a quasi-coherent $\mathcal{O}_G$ -module, and the closed image H' of f exists and is the closed subscheme of G defined by the quasi-coherent ideal

$$\mathcal{I} = \ker(\mathcal{O}_G \longrightarrow f_*(\mathcal{O}_H))$$

⁵⁸Editor's note: we have added the results that follow, taken from [Per75]. Note that Lemma 6.1 can be expressed, in Weil's language, by saying that "every point of G is a product of two generic points."

(cf. EGA I, § 9.5).

Write c_G, μ_G (respectively, c_H, μ_H) for the inversion and multiplication morphisms of G (respectively, H). Then $c_G \circ f = f \circ c_H$ factors through the closed subscheme $c_G(H')$, whence

$$H' \subset c_G(H') \quad \text{and therefore} \quad H' = c_G(H'),$$

since $c_G^2 = \text{id}_G$. Similarly,

$$f \circ \mu_H = \mu_G \circ (f \times f)$$

factors through H' , so $f \times f$ factors through the closed subscheme $\mu_G^{-1}(H')$ of $G \times G$.^{*} On the other hand, since formation of the closed image commutes with flat base change (EGA III, 1.4.15, and EGA IV₁, 1.7.21), the closed image of $f \times \text{id}_H$ (respectively, $\text{id}_{H'} \times f$) is $H' \times H$ (respectively, $H' \times H'$). Hence, by “transitivity of closed images” (EGA I, 9.5.5), the closed image of $f \times f$ is $H' \times H'$. It is therefore contained in $\mu_G^{-1}(H')$; that is, the restriction of μ_G to $H' \times H'$ factors through H' . This proves that H' is a closed group subscheme of G . Write $i : H' \hookrightarrow G$ for the inclusion.

Now $f = i \circ f'$, where $f' : H \rightarrow H'$ is schematically dominant and quasi-compact (because f is quasi-compact and i is separated). By 6.2, f' is therefore surjective. This proves 6.4.

We can now state the following theorem ([Per75], V, 3.1 and 3.2; see also [Per76], 0.0 and 0.1).

Theorem 6.5 (D. Perrin).— *Let G be a quasi-compact k -group.*

- (i) *G is the projective limit of a filtered system (G_i) of k -groups of finite type (whose transition morphisms $u_{ij} : G_j \rightarrow G_i$ are affine for i sufficiently large), and the morphisms $G \rightarrow G_i$ are faithfully flat (and affine for i sufficiently large).*
- (ii) *Let H be a closed k -subgroup of G . Then the fpqc quotient sheaf $\widetilde{G/H}$ is a k -scheme in either of the following two cases:*
 - (1) *the immersion $H \rightarrow G$ is of finite presentation; in this case G/H is of finite type over k ;*
 - (2) *H is normal in G .*

For the proof of this theorem, which rests on several intermediate theorems, we refer to [Per75]. For the reader's convenience, however, we prove the two corollaries below; cf. [Per75], V, 3.3–3.4, or [Per76], 4.2.3–4.2.5.

Corollary 6.6.— *Let $f : G \rightarrow H$ be a quasi-compact morphism of k -groups.*

- (i) *If f is schematically dominant, then it is faithfully flat.*
- (ii) *This is so, in particular, if H is affine and the morphism*

$$f^\sharp : \mathcal{O}(H) \longrightarrow \mathcal{O}(G)$$

is injective.

^{*}Translator's note: in these two occurrences the French source prints π_H and π_G , although the immediately preceding sentence defines the multiplication morphisms as μ_H and μ_G . The multiplication symbols required by the identity are used here.

Proof. Suppose that f is schematically dominant. By 6.2(i), f is surjective, so by 2.5.3 it is enough to prove that f is flat at the identity element of G . Since

$$\mathcal{O}_{H,e} = \mathcal{O}_{H^0,e}$$

(cf. 2.6.5), we may replace H by H^0 , and hence suppose that H is irreducible. Then H is quasi-compact (loc. cit.); since f is quasi-compact, G is also quasi-compact. By 6.5(i),

$$H = \varprojlim_i H_i,$$

where each H_i is an algebraic k -group. Let N_i be the kernel of $G \rightarrow H_i$; it is a closed normal k -subgroup of G . Moreover, since the identity section $\{e\} \rightarrow H_i$ is of finite presentation, the immersion $N_i \rightarrow G$ is of finite presentation. Hence, by 6.5(ii), the fpqc quotient G/N_i is an algebraic k -group. We then have a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ q_i \downarrow & & \downarrow p_i \\ G/N_i & \xrightarrow{f_i} & H_i \end{array}$$

in which f is schematically dominant and p_i, q_i are faithfully flat, hence schematically dominant. Thus

$$f_i \circ q_i = p_i \circ f$$

is schematically dominant, and therefore so is f_i . On the other hand, f_i is a monomorphism, hence a closed immersion because G/N_i and H_i are algebraic (2.5.2). It follows that f_i is an isomorphism, and consequently $G \rightarrow H_i$ is faithfully flat. By [BAC], I, § 2.7, Proposition 9, the morphism $G \rightarrow H$ is then flat; it is also surjective by 6.2(i), and is therefore faithfully flat. This proves (i). Assertion (ii) follows, because if H is affine and $f^\sharp$ is injective, then the closed image of f is H , so f is schematically dominant (cf. 6.4).

Corollary 6.7.— *Let $u : G \rightarrow H$ be a morphism of k -groups and set $N = \ker(u)$. Suppose that u is quasi-compact.*

- (i) *The fpqc quotient sheaf $\widetilde{G/N}$ is represented by a k -group scheme G/N , and u factors as*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ p \downarrow & \nearrow i & \\ G/N & & \end{array}$$

where p is faithfully flat and i is a closed immersion.

- (ii) *In particular, if u is a monomorphism, then u is a closed immersion; and if u is schematically dominant, then u is faithfully flat.*

Proof. (i) By 6.4 and 6.6, the closed image G' of u is a closed group subscheme of H , and

$$p : G \longrightarrow G'$$

is faithfully flat and quasi-compact.* Clearly $\ker(p) = \ker(u) = N$. Thus, by Exposé IV, 3.3.2.1 and 5.1.7, G' represents the fpqc quotient sheaf $\widetilde{G/N}$.

*Translator's note: the French source prints "a closed group subscheme of G " at this point. Since $u : G \rightarrow H$, its closed image lies in H , as the displayed factorization also shows.

(ii) The second assertion is contained in 6.6; let us prove the first. If u is a monomorphism, then so is p . Thus p is both a monomorphism and an effective epimorphism, and is therefore an isomorphism (cf. Exposé IV, 1.14). This proves 6.7.

Let us finally record the following corollaries (cf. [Per76], 4.2.6–4.2.8).

Corollary 6.8.— *The category of commutative quasi-compact k -group schemes is abelian.*

Taking 6.7 into account, the proof is analogous to that of 5.4.2.

Corollary 6.9.— *If $\text{char}(k) = 0$, every k -group scheme G is geometrically reduced.*

Indeed, if $g \in G$ and K is an algebraic closure of $\kappa(g)$, then

$$\mathcal{O}_{G,e} \otimes_k K \simeq \mathcal{O}_{G_K, g_K}.$$

It is therefore enough to prove that $\mathcal{O}_{G,e} = \mathcal{O}_{G^0,e}$ is geometrically reduced. We are thus reduced to the case in which G is connected, hence quasi-compact by 2.6.5. The result then follows from 6.5 and Cartier’s theorem for algebraic groups (cf. Exposé VI_B, 1.6.1, or [DG70], § II.6, Theorem 1.1).

Corollary 6.10.— *Let G be a quasi-compact k -group, and suppose that k is algebraically closed.*

- (i) *If $f : G \rightarrow H$ is a faithfully flat morphism of k -groups, then the induced map $G(k) \rightarrow H(k)$ is surjective.*
- (ii) *The set of rational points is dense in G .*

For the proof, we refer to [Per75], V, 3.7 and 3.9.

Bibliography

59

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⁵⁹Editor’s note: additional references cited in this Exposé.